

Package ‘mediaPlanR’

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Description Reproducible tools for cross-media reach and exposure-frequency planning. Provides a validated media-plan data contract; exact reach and exposure distributions under random duplication (Sainsbury and Binomial models); the Beta-Binomial accumulation model and its multi-vehicle version of Metheringham (1964) <[doi:10.1080/00218499.1964.12519751](https://doi.org/10.1080/00218499.1964.12519751)>; the ad hoc duplication and accumulation formulas of Agostini (1961) <[doi:10.1080/00218499.1961.12519620](https://doi.org/10.1080/00218499.1961.12519620)> and Hofmans (1966) <[doi:10.1177/002224376600300307](https://doi.org/10.1177/002224376600300307)>; and the multivariate exposure distribution models of Danaher (1991) <[doi:10.1177/002224379102800311](https://doi.org/10.1177/002224379102800311)> (canonical expansion), Leckenby and Rice (1986) <[doi:10.1080/00913367.1986.10673014](https://doi.org/10.1080/00913367.1986.10673014)> (Morgensztern sequential aggregation), Hyo Gyoo Kim (2005) (canonical sequential aggregation), Heejin Kim (1994) (conditional Beta distribution), and Cheong (2007) and Cheong, Leckenby and Eakin (2011) <[doi:10.2753/JOA0091-3367400101](https://doi.org/10.2753/JOA0091-3367400101)> (multivariate Beta-Binomial). Also includes target-audience metrics, an explicitly scoped Negative-Binomial count approximation, and budget-constrained optimization of effective reach that reports whether a global optimum was verified.

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URL <https://github.com/majesus/mediaPlanR>

BugReports <https://github.com/majesus/mediaPlanR/issues>

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audience_metrics	<i>Target-audience quality metrics</i>
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Description

Separates target composition, target rating and affinity. This prevents the common error of multiplying a gross audience by an affinity index and obtaining a target audience larger than the gross audience.

Usage

```
audience_metrics(
  gross_audience,
  target_audience,
  gross_universe,
  target_universe
)
```

Arguments

gross_audience	Total audience of each channel, in people, whatever the demographic profile of its members (as opposed to target_audience, the part that belongs to the target). This is not the "gross audience" that <code>calc_agostini_duplication()</code> uses for the sum of several vehicles' audiences.
target_audience	Audience belonging to the target, in people.
gross_universe	Total planning universe (the whole population), in people, as opposed to target_universe.
target_universe	Target population within the universe, in people.

Details

All inputs must be finite and logically compatible: the target audience cannot exceed the gross audience or the target universe, the gross audience cannot exceed the gross universe, and the target universe cannot exceed the gross universe. Scalars are recycled to a common length.

The affinity index compares the target's share of a channel's audience with the target's share of the universe,

$$\text{affinity} = 100 \times \frac{T/G}{T_u/G_u},$$

where T and G are the target and gross audiences and T_u and G_u the target and gross universes. Algebraically this equals the target rating divided by the gross rating, $100 \times (T/T_u)/(G/G_u)$, so the two readings of affinity always coincide and only one column is returned. An index above 100 means the channel over-delivers the target relative to the universe.

Value

A data frame with one row per channel and columns

- gross_audience and target_audience, as supplied;
- target_composition: share of the gross audience that belongs to the target, target_audience / gross_audience (NA when the gross audience is zero);

- **target_rating**: share of the target universe reached, $\text{target_audience} / \text{target_universe}$;
- **gross_rating**: share of the gross universe reached, $\text{gross_audience} / \text{gross_universe}$;
- **affinity_index**: affinity, as defined above (NA when the gross audience is zero).

Examples

```
audience_metrics(
  gross_audience = c(300000, 180000),
  target_audience = c(180000, 90000),
  gross_universe = 1000000,
  target_universe = 400000
)
```

bbd_reach_example	<i>Illustrative inputs for fitting a Beta-Binomial to an external reach</i>
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Description

List of arguments ready for `do.call()` with `fit_bbd_to_reach()`: three vehicles and an external reach. Original illustrative data, not derived from any published source.

Usage

```
bbd_reach_example
```

Format

A list with the components:

insertions Number of insertions in each vehicle (three vehicles in this example; `fit_bbd_to_reach()` accepts any number).

audiences Audience of each vehicle, in people per insertion (one value per vehicle).

reach External schedule reach, in people.

population Population size, in people.

Examples

```
data(bbd_reach_example)
do.call(fit_bbd_to_reach, bbd_reach_example)
```

beta_binomial_example *Illustrative inputs for the Beta-Binomial model*

Description

List of arguments ready for `do.call()` with `calc_beta_binomial()`. Original illustrative data, not derived from any published source.

Usage

```
beta_binomial_example
```

Format

A list with the components:

R1 Reach after the first insertion, as a proportion.

R2 Cumulative reach after the second insertion, as a proportion.

insertions Total number of planned insertions.

population Population size, in people.

Examples

```
data(beta_binomial_example)
do.call(calc_beta_binomial, beta_binomial_example)
```

calc_agostini_duplication

Reach under Agostini's duplication model

Description

Implements Agostini's (1961) ad hoc formula for the reach of a plan with one insertion in each of several vehicles when the audience of each vehicle and the audience duplicated between every pair of vehicles are known. It belongs to the "duplication" domain of Aldás Manzano's (1998) three-way split, together with `calc_hofmans_duplication()`, and estimates reach only, not the exposure distribution.

Usage

```
calc_agostini_duplication(audiences, population, duplication_matrix, k = 1.125)
```

Arguments

audiences	Numeric vector with the audience of each vehicle for one insertion, in people.
population	Population size, in people. No audience can exceed it; otherwise the function stops with an error.
duplication_matrix	Symmetric numeric matrix whose element $[i, j]$ is the number of people who are in the audience of both vehicle i and vehicle j (one insertion in each), in people (not a proportion). Diagonal values are ignored.
k	Agostini's empirical duplication coefficient. The default, 1.125, is the value Agostini fitted to a 1957 French print readership survey and found to work for French and US magazines. Other authors fitted other values to other data (Aldás Manzano, 1998, p. 156), so k should be calibrated to the media type analyzed whenever data allow it.

Details

Let $A = \sum_{i=1}^m A_i$ be the gross audience (with m the number of vehicles), also called duplicated audience: the simple sum of the vehicle audiences, in which a person who belongs to the audiences of several vehicles is counted once for each of them (Aldás Manzano, 1998, "audiencia bruta"; Kim, 2005, "gross audience"). It is not the reach, which counts each person once. Let $D = \sum_{i=1}^{m-1} \sum_{j=i+1}^m A_{ij}$ be the sum of the duplicated audiences of every pair of vehicles, each pair counted once. Agostini observed that the ratio of reach to gross audience is a function of the ratio of duplication to gross audience, $R_m/A = 1/(1 + kD/A)$, which gives

$$R_m = \frac{A^2}{A + kD}.$$

The coefficient k is constant across vehicle pairs. `calc_hofmans_duplication()` replaces it by a pairwise coefficient computed from each pair's own audiences.

The formula is empirical and can return a reach outside its logical range (below the largest audience, or above the population or the gross audience). This can happen because the duplications or k are not consistent with the formula, but also because the formula is an approximation: five independent vehicles that each reach 50% of the population, with a duplication of 25% in every pair (a valid joint distribution, with a true reach of 96.875%), give 117.6% with the default k . The value is then returned unchanged, with a warning.

Value

A list of class "reach_agostini_duplication" with components:

- reach: list with percent and people.
- k: the duplication coefficient used.
- gross_audience: sum of the vehicle audiences (duplicated people counted once per vehicle), in people.
- total_duplication: sum of the pairwise duplicated audiences, in people.
- n_vehicles: number of vehicles.

References

Agostini, J. M. (1961). How to estimate unduplicated audiences. *Journal of Advertising Research*, 1(3), 11-14. doi:10.1080/00218499.1961.12519620

Aldás Manzano, J. (1998). Modelos de determinación de la cobertura y la distribución de contactos en la planificación de medios publicitarios impresos (Models for determining reach and exposure distribution in print media planning). Doctoral dissertation, Universidad de Valencia, Spain. Section 3.2.1.1, equations 3.54-3.57.

Kim, H. G. (2005). A Canonical Sequential Aggregation Media Model. Doctoral dissertation, The University of Texas at Austin, pp. 44-45.

See Also

[calc_hofmans_duplication\(\)](#) for the pairwise refinement of this formula, and [calc_sainsbury\(\)](#) and [calc_binomial\(\)](#) for models that assume random duplication instead of observing it.

Examples

```
data(duplication_example)
result <- do.call(calc_agostini_duplication, duplication_example)
result

# A different duplication coefficient
calc_agostini_duplication(
  duplication_example$audiences, duplication_example$population,
  duplication_example$duplication_matrix, k = 1
)$reach$percent
```

calc_beta_binomial	<i>Reach and exposure distribution under the Beta-Binomial model</i>
--------------------	--

Description

Implements the Beta-Binomial model for one vehicle with several insertions. In Aldás Manzano's (1998) classification it is an *accumulation* model, like [calc_hofmans_accumulation\(\)](#): it describes how the audience of a single vehicle accumulates over its insertions (the function name omits the word, unlike the Hofmans one, because the Beta-Binomial name is the standard one in the literature and there is no second Beta-Binomial model that could be confused with it). Individuals differ in their exposure probability, which follows a Beta distribution with shape parameters alpha and beta; given that probability, the number of exposures in insertions is Binomial. Mixing the two yields the Beta-Binomial exposure distribution. The model needs only two data points, the reach after one insertion (R1) and after two insertions (R2), to estimate both shape parameters.

Usage

```
calc_beta_binomial(R1, R2, insertions, population = 1)
```

Arguments

R1	Reach after the first insertion (the vehicle's audience), as a proportion of the population, a number in (0, 1] (not a percentage). If the audience is known in people, divide it by the population.
----	--

R2	Cumulative reach after the second insertion (the proportion of the population exposed at least once in two insertions), as a proportion (not a percentage), with $R_1 \leq R_2 \leq 2 * R_1 - R_1^2$. If R2 is outside that range the function stops with an explanatory error.
insertions	Total number of planned insertions in the vehicle (a positive integer).
population	Population size, in people. It only scales the number of people reported; the default, 1, leaves people equal to the proportion.

Details

The method-of-moments estimators are

$$\hat{\alpha} = \frac{R_1(R_2 - R_1)}{2R_1 - R_1^2 - R_2}, \quad \hat{\beta} = \frac{\hat{\alpha}(1 - R_1)}{R_1}.$$

The estimators require $R_1 \leq R_2 \leq 2R_1 - R_1^2$. The two bounds are valid degenerate cases: at $R_2 = R_1$ all individuals are either always or never exposed (the *polarized* limit, $\alpha = \beta = 0$), and at $R_2 = 2R_1 - R_1^2$ exposures are independent (the *binomial* limit, $\alpha = \beta = \text{Inf}$). Both limits are handled explicitly.

Value

A list of class "reach_beta_binomial" with components:

- reach: list with percent and people.
- distribution: list with percent and people, for one to insertions exposures.
- cumulative: list with percent and people, for individuals exposed at least once, at least twice, and so on.
- population: the population size used (people).
- population: the population size used, in people.
- parameters: list with alpha, beta, mean_probability (the mean of the Beta distribution, R1), zero_contact_probability (the percentage of the population with no exposure, on a 0-100 scale unlike the probability columns of other results) and type ("beta_binomial", "binomial_limit" or "polarized_limit").

References

Aldás Manzano, J. (1998). Modelos de determinación de la cobertura y la distribución de contactos en la planificación de medios publicitarios impresos (Models for determining reach and exposure distribution in print media planning). Doctoral dissertation, Universidad de Valencia, Spain. Section 3.1.2.5, equations 3.22-3.28, with the parameter estimators on page 134.

See Also

[calc_sainsbury\(\)](#), [calc_binomial\(\)](#) and [calc_metheringham\(\)](#) for plans with several vehicles, [calc_hofmans_accumulation\(\)](#) for an ad hoc accumulation model, and [nbd_exposure_distribution\(\)](#) for a Negative-Binomial count approximation, which applies to unbounded count processes rather than to a finite number of insertions.

Examples

```
result <- calc_beta_binomial(R1 = 0.50, R2 = 0.55, insertions = 5,
                             population = 1000000)

result$reach$percent
result$parameters$alpha
result$parameters$beta
```

calc_binomial

Reach and exposure distribution under the Binomial model

Description

Implements the Binomial model of Lee and Burkhart (1960), reviewed by Chandon, to calculate reach and the exposure distribution (and its cumulative counterpart) of a media plan with several vehicles and one or more insertions in each of them (argument insertions). The model assumes random duplication *and* random accumulation (repeat insertions in the same vehicle are also independent), homogeneous individuals and homogeneous vehicles. Under these assumptions every insertion is an independent Bernoulli trial with the same exposure probability, which is the insertion-weighted mean audience of the plan (each vehicle's audience counted once per insertion) divided by the population. Exposure probabilities are assumed stationary over time.

Usage

```
calc_binomial(audiences, population, insertions = rep(1L, length(audiences)))
```

Arguments

audiences	Numeric vector with the audience of each vehicle, in people per insertion.
population	Population size, in people. No audience can exceed it; otherwise the function stops with an error.
insertions	Non-negative integer vector with the number of insertions planned in each vehicle. It defaults to one insertion per vehicle, the model's original scope.

Details

Aldás Manzano (1998, Section 3.3.1.1) defines the exposure probability as $p = \bar{A}/P$, where $\bar{A}$ is the simple mean of the vehicle audiences and every vehicle receives the same number of insertions n , so a plan with m vehicles has $N = nm$ insertions. With unequal insertion counts, this function uses the insertion-weighted mean audience, $\sum_{i=1}^m n_i A_i / \sum_{i=1}^m n_i$. The two definitions coincide when all vehicles receive the same number of insertions, and the weighted mean preserves the plan's expected number of exposures in general.

The number of exposures then follows a Binomial distribution with N trials and success probability p , and reach equals $1 - (1 - p)^N$.

Value

A list of class "reach_binomial" with components:

- reach: list with percent and people.
- distribution: list with percent and people, one value for each number of exposures from one to the total number of insertions.
- cumulative: list with percent and people, for individuals exposed at least once, at least twice, and so on.
- mean_probability: exposure probability used for every insertion.

References

Aldás Manzano, J. (1998). Modelos de determinación de la cobertura y la distribución de contactos en la planificación de medios publicitarios impresos (Models for determining reach and exposure distribution in print media planning). Doctoral dissertation, Universidad de Valencia, Spain. Section 3.2.2.1 for one insertion per vehicle; Section 3.3.1.1 for several insertions.

Lee, A. M., & Burkhart, A. J. (1960). Some optimization problems in advertising media planning. *Operational Research Quarterly*, 11(3), 113-122.

Chandon, J.-L. (1976). A comparative study of media exposure models. Unpublished doctoral dissertation, Northwestern University, Evanston, IL. (Aldás Manzano, 1998, cites Chandon's work as Chandon, 1985.)

See Also

[calc_sainsbury\(\)](#) for heterogeneous vehicles, [calc_beta_binomial\(\)](#) and [calc_metheringham\(\)](#) for models with heterogeneous individuals, and [estimate_reach\(\)](#) to run this model from a [media_plan\(\)](#) object.

Examples

```
audiences <- c(300000, 400000, 200000)
population <- 1000000
result <- calc_binomial(audiences, population)
result$reach$percent
result$mean_probability

# The same three vehicles with several insertions each
calc_binomial(audiences, population, insertions = c(4, 6, 10))
```

calc_canex

Canonical Expansion model (CANEX)

Description

Implements Danaher's (1991) canonical expansion model for the reach and exposure distribution of a multi-vehicle schedule. Each vehicle's exposure distribution is a Beta-Binomial, which accounts for the heterogeneity of individuals, and the observed duplication between vehicles enters through a second-order correlation term.

Usage

```
calc_canex(vehicles_data, duplications, population = 1)
```

Arguments

vehicles_data	Data frame with one row per vehicle and columns • insertions: number of insertions in the vehicle (a positive integer). The column may also be named k, Danaher's notation, so that the same data frame can be used by <code>calc_csd()</code>, <code>calc_msad()</code>, <code>calc_cbd()</code> and <code>calc_mbd()</code>, which name it insertions; • R1: reach after the first insertion, as a proportion between 0 and 1 (not a percentage), in (0, 1]; • R2: cumulative reach after the second insertion (the proportion exposed at least once in two insertions), a proportion between R1 and the independence limit $2 * R1 - R1^2$. If R2 is outside that range the function stops with an explanatory error.
duplications	Symmetric matrix whose element [i, j] is the proportion of the population that is exposed to both vehicle i and vehicle j (one insertion in each), a number between 0 and 1 (not a percentage). It is not the proportion exposed to each vehicle separately. The diagonal is ignored.
population	Number of people in the population (a count, not a proportion). It only converts proportions into people: the people columns of the result are the probability times population, and with the default, 1, they equal the probability.

Details

The model follows these steps:

1. It computes each vehicle's Beta-Binomial parameters from R1 and R2 by the method of moments.
2. It turns the duplications into canonical correlations, $r_{ij} = (p_{ij} - p_i p_j) / \sqrt{p_i(1 - p_i)p_j(1 - p_j)}$, with p_i the one-insertion reach of vehicle i and p_{ij} the duplication of vehicles i and j : the proportion of the population exposed to both of them.
3. It evaluates, for every combination of exposure levels, the joint probability $\prod_{i=1}^m f_i(x_i) \{1 + \sum_{i=1}^{m-1} \sum_{j=i+1}^m r_{ij} (x_i - \mu_i)(x_j - \mu_j) / (\sigma_i \sigma_j)\}$, where m is the number of vehicles, the double sum runs over every pair of vehicles, and f_i , μ_i and σ_i^2 are the Beta-Binomial probability function, mean and variance of vehicle i . Only second-order terms are kept.
4. It sets to zero the negative probabilities that the truncated expansion can produce, renormalizes the joint distribution to sum to one, and sums it over the total number of exposures.

The number of combinations, $\prod_{i=1}^m (n_i + 1)$ with n_i the insertions of vehicle i , grows exponentially with the number of vehicles and insertions. The model is therefore intended for small and medium schedules, and the function stops with an informative error above 2,000,000 combinations instead of exhausting memory.

Value

A list of class "reach_canex" with components:

- population: the population size used.
- reach: list with probability, percent and people.

- `average_frequency`: average number of exposures among the people reached.
- `distribution`: data frame with contacts (total exposures, from zero), probability, percent, people and `cumulative_probability` (the probability of contacts or more exposures).
- `diagnostics`: list with `negative_mass_truncated` (probability mass that was negative before truncation), `mass_before_renormalization` and `correlation_min_eigenvalue` (the smallest eigenvalue of the correlation matrix), `mean_exposures_expected` (the mean number of exposures per person of the population: the sum over vehicles of insertions times R1, preserved by the expansion) and `mean_exposures_result` (the mean of the returned distribution). They show how much the second-order approximation had to be corrected.

Truncation of negative probabilities

Danaher (1991, pp. 363-364) shows that the probabilities of the expansion sum to one and reports, in his application, negative probabilities in a small fraction of the cells, with a mean magnitude of the order of $1e-7$, which he sets to zero. That practice rests on the negative mass being negligible. This function applies it at any size and reports the truncated mass in `diagnostics$negative_mass_truncated`, together with the mean number of exposures per person that the expansion preserves (`mean_exposures_expected`, the sum over the vehicles of insertions times R1) and the one of the returned distribution (`mean_exposures_result`). Both are averages over the whole population, including the people never exposed, so they are not the average frequency among the people reached. When the truncated mass is not negligible, truncating and renormalizing can change the mean. The function then issues a warning of class `mediaPlanR_canex_truncation`; the threshold, a total negative mass of $1e-4$, is a decision of this package and not of Danaher (1991). For example, two vehicles with two insertions each, $R1 = c(0.3, 0.2)$, $R2 = c(0.4, 0.3)$ and zero duplication have an expected mean of 1 exposure per person (2 insertions $\times$ 0.3 plus 2 insertions $\times$ 0.2, counted over the whole population) and a negative mass of 0.0076, and the truncated distribution has a mean of 1.0227.

Domain of validity

In plain words. These models take the overlaps between vehicles (the duplications) from your data and use them to build the whole distribution of exposures. For that to be possible the overlaps you supply must be possible, and plausible together:

- Two vehicles cannot overlap more than the smaller of their audiences, nor less than their audiences force them to (two audiences of 60% of the population must overlap by at least 20%). If a duplication breaks these limits, the function stops with an error naming the pair: the data cannot be right and should be reviewed.
- The overlaps of three vehicles must also fit together. If they cannot, the function stops with an error naming the three vehicles.
- The models were built for the usual situation in which vehicles overlap at least as much as chance alone would produce, that is, at least the product of their audiences (two vehicles with audiences of 30% and 20% overlap by 6% if people chose vehicles independently). When a pair overlaps clearly less than that, the results can differ a lot from one model to another, and the function issues a warning that names the pair.

An error means that the inputs are impossible: correct them. A warning does not invalidate the calculation, but it asks for caution: check the duplication of the pair named, which is often a measurement problem (see below), and compare the result with another model. The rest of this section gives the technical detail.

The multivariate models (`calc_canex()`, `calc_csd()`, `calc_msad()`, `calc_cbd()` and `calc_mbd()`) take the observed one-insertion duplications as given and check them against the Fréchet bounds

implied by the audiences and, for every triple of vehicles, against joint feasibility. The published methods have three limits that follow from this.

First, Hong (1998, p. 238) notes that the models he evaluates assume a duplication between vehicles not below the random one, the product of the audiences, and treats null observed duplications as measurement error. With a lower duplication the models can diverge considerably. The functions warn (class `mediaPlanR_low_duplication`) when the canonical correlation of a pair is below -0.1. That threshold is a decision of this package: the published reference plans (Kim 2005, Appendix B, and Hong 1998, Appendix E; 80 plans) include pairs slightly below the random duplication, with canonical correlations down to -0.071, and they raise no warning. For two vehicles with two insertions each, $R1 = 0.30$ and $R2 = 0.45$, a duplication of zero gives a reach of 0.90 under MSAD, 0.81 under MBD, 0.78 under CSD and CBD, and 0.79 under CANEX. Danaher's canonical expansion itself admits negative correlations. Hong (1998, pp. 238-239) treats a null duplication as a measurement error and replaces it by $\hat{p}_{ij} = K' p_i p_j$, where K' is the ratio between observed and random duplication over the pairs with non-zero observed duplication. The package does not implement this imputation: `calc_agostini_duplication()` and `calc_hofmans_duplication()` estimate reach from a duplication matrix and do not impute its elements. If duplications are imputed beforehand, the result must be checked against the bounds described below. A null duplication is not always a measurement error: it can also be a genuine observation.

Second, pairwise duplications that satisfy the Fréchet bounds can still be impossible jointly for three or more vehicles. For every triple of vehicles the functions check that the exposure to all three can have a non-negative probability, given the audiences and the three pairwise duplications, and stop with an error when it cannot. That condition is necessary but not sufficient for four or more vehicles, for which the published methods check feasibility at most to first order (Cheong 2007); `calc_mbd()` and `calc_cbd()` zero any negative cell that remains, and `calc_canex()` truncates it.

Third, a low duplication between vehicles with several insertions can make the target of a sequential model invalid. For two vehicles with two insertions each, $R1 = 0.5$ and $R2 = 0.75$, `calc_csd()` and `calc_cbd()` stop with a duplication of zero, and `calc_msad()` with a duplication of zero or 0.5, instead of altering an invalid target.

References

- Danaher, P. J. (1991). A canonical expansion model for multivariate media exposure distributions: A generalization of the "duplication of viewing law". *Journal of Marketing Research*, 28(3), 361-367. doi:10.1177/002224379102800311
- Kim, H. G. (2005). A Canonical Sequential Aggregation Media Model. Doctoral dissertation, The University of Texas at Austin, pp. 56-58.

See Also

`calc_beta_binomial()` for the univariate model (one vehicle), and `calc_csd()` and `calc_cbd()`, whose between-vehicle step reuses this canonical expansion.

Examples

```
# Two vehicles, two insertions each
data(canex_example)
result <- do.call(calc_canex, canex_example)
result
result$diagnostics

# Inputs of the first two vehicles of Kim's (2005) worked example
# (Tables 4.2.2.1 and 4.2.2.2), with a population of 500,000 people
vehicles <- data.frame(
```

```

insertions = c(2, 2),
R1 = c(0.4902, 0.0333),
R2 = c(0.5805, 0.0502)
)
duplications <- matrix(c(NA, 0.0157, 0.0157, NA), nrow = 2)
calc_canex(vehicles, duplications, population = 500000)$reach

```

calc_cbd

Conditional Beta Distribution model

Description

Implements the Conditional Beta Distribution (CBD) of Kim (1994), reviewed in Kim (2005), for several vehicles with several insertions each. The "(0,1) grid" is the set of the 2^m combinations of being exposed (1) or not exposed (0) to the first insertion of each of the m vehicles; the notation "(0,1)" refers to those two states, not to an interval. Between-vehicle duplication is modeled first, at the one-insertion (0,1) level, by Danaher's (1991) second-order canonical expansion – the mechanism `calc_canex()` and `calc_csd()` use. Conditionally on the (0,1) exposure state of every vehicle, the remaining insertions of each vehicle follow a Beta-Binomial distribution whose Beta parameters are updated by that state, and the vehicles are independent given the grid (Danaher 1992a). The zero cell of the grid is then conformed to the canonical expansion of all the insertions.

Usage

```

calc_cbd(
  vehicles_data,
  duplications,
  aggregation_order = c("audience_desc", "given"),
  population = 1,
  tolerance = 1e-08
)

```

Arguments

- | | |
|-------------------|--|
| vehicles_data | Data frame with columns insertions, R1 and R2, with the same convention as <code>calc_csd()</code> (R1 and R2 are proportions between 0 and 1, not percentages, R1 is strictly between zero and one, $R1 \leq R2 \leq 2 * R1 - R1^2$, and R2 may be NA only when insertions is one; the function stops with an explanatory error if R2 is outside that range). At most 12 vehicles are supported. |
| duplications | Symmetric matrix whose element $[i, j]$ is the proportion of the population that is exposed to both vehicle i and vehicle j (one insertion in each), a number between 0 and 1 (not a percentage). It is not the proportion exposed to each vehicle separately. The diagonal is ignored. |
| aggregation_order | Either "audience_desc" (vehicles in decreasing order of one-insertion reach R1), "given" (the row order), or a permutation of the row indices. CBD does not use it: in Kim's specification the vehicles are conditionally independent given the (0,1) grid, so neither the reach nor the distribution depends on the order. The argument is accepted, validated and reported only so that the same call, and the same example data (for instance <code>csd_kim2005</code>), work for <code>calc_csd()</code> , <code>calc_msad()</code> , <code>calc_cbd()</code> and <code>calc_mbd()</code> . |

population	Number of people in the population (a count, not a proportion). It only converts probabilities into people: the people columns of the result are the probability times population, and with the default, 1, they equal the probability.
tolerance	Positive numerical tolerance for probability constraints. The Fréchet and triple-feasibility checks apply it relative to the smaller audience involved.

Details

The model follows Kim (1994, pp. 115-124):

1. The joint distribution of exposure to one insertion of each vehicle (the 2^m cells of the (0,1) grid) is obtained from the second-order canonical expansion with Bernoulli marginals, from the pairwise correlations implied by the observed one-insertion duplications.
2. Given that vehicle i was ($x_i = 1$) or was not ($x_i = 0$) exposed to its first insertion, its remaining $n_i - 1$ insertions follow a Beta-Binomial distribution with parameters $(\alpha + x_i, \beta + 1 - x_i)$, where α and β are fitted from R1 and R2. The vehicle's total exposure is that distribution shifted by x_i . The distributions of the vehicles are convolved within each cell of the grid and weighted by the cell probability. A vehicle with one insertion contributes only its state x_i .
3. In the all-zero cell of the grid, the probability of zero exposures is set to the canonical expansion of the zero cell computed with all the insertions (Danaher 1991, as in `calc_canex()`), and the remaining levels of that cell are rescaled so that the cell keeps its total probability (Kim 1994, p. 124).

With a binomial-limit vehicle (R2 equal to the reach under independence) the conditional distribution is the binomial one, and with a polarized one (R2 equal to R1) it is the all-or-nothing exposure; both are the limits of the Beta-Binomial updating.

Unlike `calc_mbd()`, the between-vehicle step of CBD needs no imputation for three or more vehicles. The canonical expansion uses the mean R_{1i} and variance $R_{1i}(1 - R_{1i})$ of each vehicle's one-insertion exposure; this guarantees that summing out all other vehicles returns each vehicle's own Bernoulli marginal exactly, and with zero correlation the model reduces to the exact convolution of the vehicles' Beta-Binomial marginals. With correlation, the final zero-cell adjustment does not preserve the mean number of exposures of each vehicle exactly; the departure is reported in `diagnostics$mean_error` and was between 0.0003 and 0.013 exposures in the package's reference schedules.

Value

A `reach_cbd` object: a list with `reach` (probability, percent and people), `average_frequency`, the complete exposure distribution (`contacts`, `probability`, `percent`, `people` and `cumulative_probability`), the `aggregation_order` and `aggregation_rule`, the within-vehicle `vehicle_expansion` (the distribution used for each vehicle) and `diagnostics`, which include the canonical zero-exposure probability, the zero-cell rescaling factor and whether the negative-probability safety net of `calc_mbd()` had to be engaged (`negative_mass_adjusted`, `cells_adjusted`).

Validation against published results

The implementation reproduces the CBD columns of the two-vehicle comparisons of Kim (2005, Appendix B) and of Hong (1998, Appendix E) within the rounding of those sources (0.01 percentage points), taking the duplication that each source implies, and the worked example of Kim (1994, pp. 125-139; three vehicles with 2, 1 and 3 insertions, SMRB 1979 data) except for its zero cell. In that example the zero of the canonical expansion is printed as 0.5066, whereas the formula applied to the printed inputs gives about 0.510 (and no choice of the inputs within their rounding goes below 0.507), so the printed value is an arithmetic error of the source; this function evaluates the formula,

so its zero and one-contact probabilities differ from the printed ones by 0.36 and 0.29 percentage points in that example. See the package tests for the exact cases.

Because the canonical expansion can assign small negative probabilities to some (0,1) cells – the limitation documented for `calc_canex()` – any cell that is still negative at the end is set to zero and that mass is redistributed proportionally, as in the final safety net of `calc_mbd()`. This step is an extension of the package, not part of Kim’s specification, and is reported in diagnostics. If the canonical zero-exposure probability is not a probability, the function stops, as `calc_csd()` does.

The number of cells of the (0,1) grid grows as 2^m , so the number of vehicles is limited to 12.

Domain of validity

In plain words. These models take the overlaps between vehicles (the duplications) from your data and use them to build the whole distribution of exposures. For that to be possible the overlaps you supply must be possible, and plausible together:

- Two vehicles cannot overlap more than the smaller of their audiences, nor less than their audiences force them to (two audiences of 60% of the population must overlap by at least 20%). If a duplication breaks these limits, the function stops with an error naming the pair: the data cannot be right and should be reviewed.
- The overlaps of three vehicles must also fit together. If they cannot, the function stops with an error naming the three vehicles.
- The models were built for the usual situation in which vehicles overlap at least as much as chance alone would produce, that is, at least the product of their audiences (two vehicles with audiences of 30% and 20% overlap by 6% if people chose vehicles independently). When a pair overlaps clearly less than that, the results can differ a lot from one model to another, and the function issues a warning that names the pair.

An error means that the inputs are impossible: correct them. A warning does not invalidate the calculation, but it asks for caution: check the duplication of the pair named, which is often a measurement problem (see below), and compare the result with another model. The rest of this section gives the technical detail.

The multivariate models (`calc_canex()`, `calc_csd()`, `calc_msad()`, `calc_cbd()` and `calc_mbd()`) take the observed one-insertion duplications as given and check them against the Fréchet bounds implied by the audiences and, for every triple of vehicles, against joint feasibility. The published methods have three limits that follow from this.

First, Hong (1998, p. 238) notes that the models he evaluates assume a duplication between vehicles not below the random one, the product of the audiences, and treats null observed duplications as measurement error. With a lower duplication the models can diverge considerably. The functions warn (class `mediaPlanR_low_duplication`) when the canonical correlation of a pair is below -0.1. That threshold is a decision of this package: the published reference plans (Kim 2005, Appendix B, and Hong 1998, Appendix E; 80 plans) include pairs slightly below the random duplication, with canonical correlations down to -0.071, and they raise no warning. For two vehicles with two insertions each, $R1 = 0.30$ and $R2 = 0.45$, a duplication of zero gives a reach of 0.90 under MSAD, 0.81 under MBD, 0.78 under CSD and CBD, and 0.79 under CANEX. Danaher’s canonical expansion itself admits negative correlations. Hong (1998, pp. 238-239) treats a null duplication as a measurement error and replaces it by $\hat{p}_{ij} = K' p_i p_j$, where K' is the ratio between observed and random duplication over the pairs with non-zero observed duplication. The package does not implement this imputation: `calc_agostini_duplication()` and `calc_hofmans_duplication()` estimate reach from a duplication matrix and do not impute its elements. If duplications are imputed beforehand, the result must be checked against the bounds described below. A null duplication is not always a measurement error: it can also be a genuine observation.

Second, pairwise duplications that satisfy the Fréchet bounds can still be impossible jointly for three or more vehicles. For every triple of vehicles the functions check that the exposure to all three can have a non-negative probability, given the audiences and the three pairwise duplications, and stop with an error when it cannot. That condition is necessary but not sufficient for four or more vehicles, for which the published methods check feasibility at most to first order (Cheong 2007); `calc_mbd()` and `calc_cbd()` zero any negative cell that remains, and `calc_canex()` truncates it.

Third, a low duplication between vehicles with several insertions can make the target of a sequential model invalid. For two vehicles with two insertions each, $R1 = 0.5$ and $R2 = 0.75$, `calc_csd()` and `calc_cbd()` stop with a duplication of zero, and `calc_msad()` with a duplication of zero or 0.5, instead of altering an invalid target.

References

- Kim, H. (1994). A conditional beta distribution model for advertising media reach/frequency estimation. Unpublished doctoral dissertation, The University of Texas at Austin, pp. 115-139.
- Kim, H. G. (2005). A Canonical Sequential Aggregation Media Model. Doctoral dissertation, The University of Texas at Austin, pp. 59-64.
- Hong, J. (1998). Advertising media models for Internet reach/frequency estimation. Unpublished doctoral dissertation, The University of Texas at Austin, Appendix E.
- Danaher, P. J. (1991). A canonical expansion model for multivariate media exposure distributions: A generalization of the "duplication of viewing law". *Journal of Marketing Research*, 28(3), 361-367. doi:10.1177/002224379102800311
- Danaher, P. J. (1992a). A Markov-chain model for multivariate magazine-exposure distributions. *Journal of Business & Economic Statistics*, 10(4), 401-407. doi:10.1080/07350015.1992.10509915

See Also

`calc_mbd()` for a model with an imputed between-vehicle step, and `calc_canex()` and `calc_csd()` for the canonical expansion that this model's first and last steps reuse.

Examples

```
# Same three-vehicle inputs as the CSD example of Kim (2005), so that CBD
# can be compared with CSD and CANEX on identical data
data(csd_kim2005)
result <- do.call(calc_cbd, csd_kim2005)
result$reach
result$distribution
```

calc_csd

Canonical Sequential Aggregation Distribution model

Description

Implements Kim's (2005) Canonical Sequential Aggregation Distribution (CSD): the Beta-Binomial marginal of each vehicle is combined sequentially with the vehicles already aggregated, while the reach at every step comes from Danaher's second-order canonical expansion. The non-random convolution preserves both marginal distributions and sets the zero-exposure cell to one minus the canonical reach.

Usage

```
calc_csd(
  vehicles_data,
  duplications,
  aggregation_order = c("audience_desc", "given"),
  population = 1,
  tolerance = 1e-10
)
```

Arguments

- | | |
|-------------------|---|
| vehicles_data | Data frame with one row per vehicle (at least two rows) and columns insertions (planned insertions, a positive integer), R1 and R2. R1 is the reach after one insertion, a proportion between 0 and 1 (not a percentage) and strictly between zero and one, and R2 the cumulative reach after two insertions (the proportion exposed at least once in two insertions), a proportion with $R1 \leq R2 \leq 2 * R1 - R1^2$ (the upper limit is the reach under independent insertions). If R2 is outside that range the function stops with an explanatory error. R2 may be NA only when insertions is one. |
| duplications | Symmetric matrix whose element $[i, j]$ is the proportion of the population that is exposed to both vehicle i and vehicle j (one insertion in each), a number between 0 and 1 (not a percentage). It is not the proportion exposed to each vehicle separately. The diagonal is ignored. |
| aggregation_order | Either "audience_desc" (vehicles in decreasing order of one-insertion reach R1), "given" (the row order), or a permutation of the row indices. The vehicles are combined one at a time in this order. The order does not change the reach of the whole schedule but can change the exposure distribution. Use an explicit permutation to reproduce a published aggregation order, such as the one of Kim's worked example (labeled "TD forward" in the dissertation: aggregation guided by the total duplication of each pair, between- plus within-vehicle, in forward order). |
| population | Number of people in the population (a count, not a proportion). It only converts probabilities into people: the people columns of the result are the probability times population, and with the default, 1, they equal the probability. |
| tolerance | Positive numerical tolerance for probability constraints. The Fréchet and triple-feasibility checks apply it relative to the smaller audience involved. |

Details

For a subset of m vehicles, CSD estimates the all-zero probability as

$$P(0, \dots, 0) = \prod_{i=1}^m f_i(0) \left[1 + \sum_{i=1}^{m-1} \sum_{j=i+1}^m \rho_{ij} \frac{(0 - \mu_i)(0 - \mu_j)}{\sigma_i \sigma_j} \right],$$

where f_i , μ_i and σ_i are the Beta-Binomial probability function, mean and standard deviation of vehicle i , and ρ_{ij} is the canonical correlation obtained from the observed one-insertion duplication. Reach is one minus this probability.

The canonical expansion is a second-order approximation. Unlike `calc_canex()`, which truncates negative probabilities and renormalizes the full joint grid, `calc_csd()` does not alter an invalid

canonical target: it stops if the resulting reach is not a probability or is incompatible with the two margins being combined. This makes approximation failure visible to the analyst.

Kim's worked example rounds intermediate values to four decimals. Exact calculations from the published inputs therefore differ by a few hundredths of a percentage point from some displayed cells; the unrounded calculation is returned.

Value

A `reach_csd` object: a list with `reach` (probability, percent and people), `average_frequency`, the complete exposure distribution (`contacts`, `probability`, `percent`, `people` and `cumulative_probability`), the `vehicle_marginals`, the one-insertion `vehicle_reach`, the canonical `correlation_matrix`, the `aggregation_order` and `aggregation_rule`, the aggregation steps and numerical diagnostics. The diagnostics include `gross_mean_contacts`, the mean number of exposures per person that the inputs imply (the sum over vehicles of insertions times $R1$, averaged over the whole population), `distribution_mean_contacts`, the same mean computed from the returned distribution, and `mean_error`, their difference.

Domain of validity

In plain words. These models take the overlaps between vehicles (the duplications) from your data and use them to build the whole distribution of exposures. For that to be possible the overlaps you supply must be possible, and plausible together:

- Two vehicles cannot overlap more than the smaller of their audiences, nor less than their audiences force them to (two audiences of 60% of the population must overlap by at least 20%). If a duplication breaks these limits, the function stops with an error naming the pair: the data cannot be right and should be reviewed.
- The overlaps of three vehicles must also fit together. If they cannot, the function stops with an error naming the three vehicles.
- The models were built for the usual situation in which vehicles overlap at least as much as chance alone would produce, that is, at least the product of their audiences (two vehicles with audiences of 30% and 20% overlap by 6% if people chose vehicles independently). When a pair overlaps clearly less than that, the results can differ a lot from one model to another, and the function issues a warning that names the pair.

An error means that the inputs are impossible: correct them. A warning does not invalidate the calculation, but it asks for caution: check the duplication of the pair named, which is often a measurement problem (see below), and compare the result with another model. The rest of this section gives the technical detail.

The multivariate models (`calc_canex()`, `calc_csd()`, `calc_msad()`, `calc_cbd()` and `calc_mbd()`) take the observed one-insertion duplications as given and check them against the Fréchet bounds implied by the audiences and, for every triple of vehicles, against joint feasibility. The published methods have three limits that follow from this.

First, Hong (1998, p. 238) notes that the models he evaluates assume a duplication between vehicles not below the random one, the product of the audiences, and treats null observed duplications as measurement error. With a lower duplication the models can diverge considerably. The functions warn (class `mediaPlanR_low_duplication`) when the canonical correlation of a pair is below -0.1. That threshold is a decision of this package: the published reference plans (Kim 2005, Appendix B, and Hong 1998, Appendix E; 80 plans) include pairs slightly below the random duplication, with canonical correlations down to -0.071, and they raise no warning. For two vehicles with two insertions each, $R1 = 0.30$ and $R2 = 0.45$, a duplication of zero gives a reach of 0.90 under MSAD, 0.81 under MBD, 0.78 under CSD and CBD, and 0.79 under CANEX. Danaher's canonical

expansion itself admits negative correlations. Hong (1998, pp. 238-239) treats a null duplication as a measurement error and replaces it by $\hat{p}_{ij} = K' p_i p_j$, where K' is the ratio between observed and random duplication over the pairs with non-zero observed duplication. The package does not implement this imputation: `calc_agostini_duplication()` and `calc_hofmans_duplication()` estimate reach from a duplication matrix and do not impute its elements. If duplications are imputed beforehand, the result must be checked against the bounds described below. A null duplication is not always a measurement error: it can also be a genuine observation.

Second, pairwise duplications that satisfy the Fréchet bounds can still be impossible jointly for three or more vehicles. For every triple of vehicles the functions check that the exposure to all three can have a non-negative probability, given the audiences and the three pairwise duplications, and stop with an error when it cannot. That condition is necessary but not sufficient for four or more vehicles, for which the published methods check feasibility at most to first order (Cheong 2007); `calc_mbd()` and `calc_cbd()` zero any negative cell that remains, and `calc_canex()` truncates it.

Third, a low duplication between vehicles with several insertions can make the target of a sequential model invalid. For two vehicles with two insertions each, $R1 = 0.5$ and $R2 = 0.75$, `calc_csd()` and `calc_cbd()` stop with a duplication of zero, and `calc_msad()` with a duplication of zero or 0.5, instead of altering an invalid target.

References

- Danaher, P. J. (1991). A canonical expansion model for multivariate media exposure distributions: A generalization of the duplication of viewing law. *Journal of Marketing Research*, 28(3), 361-367. doi:[10.1177/002224379102800311](https://doi.org/10.1177/002224379102800311)
- Kim, H. G. (2005). A Canonical Sequential Aggregation Media Model. Doctoral dissertation, The University of Texas at Austin, pp. 78-97.

See Also

`calc_canex()` for the full-grid canonical expansion, `calc_msad()` for the Morgensztern sequential alternative and `calc_cbd()` for a different sequential architecture built on the same canonical expansion at the (0,1) level.

Examples

```
# Kim (2005), Tables 4.2.2.1-4.2.2.10: the order of Kim's worked example
data(csd_kim2005)
result <- do.call(calc_csd, csd_kim2005)
result$reach
result$distribution
```

calc_hofmans_accumulation

Cumulative reach under Hofmans' accumulation model

Description

Implements Hofmans' (1966) accumulation model for the cumulative reach (the proportion of the population exposed at least once) of several insertions in one vehicle – the "accumulation" domain in Aldás Manzano's (1998) three-way split. It adapts Agostini's duplication formula by replacing the number of vehicles with the number of insertions, and it needs only the audience after one and

two insertions. It is unrelated to `calc_hofmans_duplication()`, the same author's separate model for several vehicles with one insertion each.

Usage

```
calc_hofmans_accumulation(R1, R2, insertions, R3 = NULL, show_steps = FALSE)
```

Arguments

R1	Reach after the first insertion, as a proportion between 0 and 1 (not a percentage), in (0, 1].
R2	Cumulative reach after the second insertion, as a proportion (not a percentage) with $R1 < R2 < 2 * R1$ and $R2 \leq 1$.
insertions	Number of insertions in the vehicle up to which the cumulative reach is computed (N in the formulas), an integer of at least 2 (at least 3 when R3 is supplied).
R3	Optional cumulative reach after the third insertion, as a proportion (not a percentage) with $R2 < R3 < 3 * R1$ and $R3 \leq 1$. It is observed data, like R1 and R2, not something the function estimates: when supplied, the exponent alpha of Hofmans' variable coefficient is estimated from it, so that the curve reproduces R3 (see Details).
show_steps	Logical. If TRUE, prints the parameters k, d and alpha and the table of cumulative reach.

Details

Let $d = 2R_1 - R_2$ be the duplicated reach of two insertions (the proportion of the population exposed to both), assumed constant for every pair of insertions, and $k = 2R_1/R_2$ the coefficient that makes the formula reproduce R_2 exactly. Cumulative reach after N insertions is

$$R_N = \frac{(NR_1)^2}{NR_1 + k_N d \binom{N}{2}}.$$

With a constant coefficient, $k_N = k$ (equation 3.11 of Aldás Manzano, 1998). Hofmans showed that the coefficient is not constant and proposed $k_N = k(N - 1)^{\alpha-1}$ (equation 3.12), where the exponent is estimated from a third observation,

$$\hat{\alpha} = \log \left[\frac{(3R_1 - R_3)R_2}{(2R_1 - R_2)R_3} \right] / \log 2.$$

R3 therefore selects the model: without it, alpha = 1 and the constant coefficient formula applies; with it, alpha is estimated and the curve reproduces R3 exactly.

The model assumes a constant reach per insertion (R1) and a constant duplication between every pair of insertions.

Valid R1 and R2 do not guarantee a valid curve: the extrapolation can decrease, exceed 1 or exceed NR_1 . For example, $R1 = 0.60$, $R2 = 0.84$ and $N = 4$ give a cumulative reach of 1.05 at $N = 4$. The curve is then returned unchanged, with a warning.

Value

A list of class "reach_hofmans_accumulation" with components:

- **results**: data frame with insertions (number of insertions) and RN (cumulative reach, as a proportion).

- parameters: list with k, d, alpha and R3 (NA when not supplied).
- plot: a ggplot2 object showing the evolution of cumulative reach, or NULL when the suggested package ggplot2 is not installed.

References

Aldás Manzano, J. (1998). Modelos de determinación de la cobertura y la distribución de contactos en la planificación de medios publicitarios impresos (Models for determining reach and exposure distribution in print media planning). Doctoral dissertation, Universidad de Valencia, Spain. Section 3.1.1.4, equations 3.7-3.12.

Hofmans, P. (1966). Measuring the cumulative net coverage of any combination of media. *Journal of Marketing Research*, 3(3), 269-278. doi:[10.1177/002224376600300307](https://doi.org/10.1177/002224376600300307)

See Also

[calc_beta_binomial\(\)](#) for a stochastic accumulation model that also returns the exposure distribution, and [calc_hofmans_duplication\(\)](#) for the same author's model for several vehicles.

Examples

```
# Constant-coefficient model (equation 3.11)
result <- calc_hofmans_accumulation(R1 = 0.06, R2 = 0.103, insertions = 5)
result$results
result$parameters

# With an observed third insertion, Hofmans' exponent is estimated
calc_hofmans_accumulation(R1 = 0.06, R2 = 0.103, insertions = 5,
                          R3 = 0.135)$parameters
```

calc_hofmans_duplication

Reach under Hofmans' duplication model

Description

Implements Hofmans' (1966) *duplication* model: an ad hoc correction of Agostini's (1961) formula for several vehicles with one insertion each – the same "duplication" domain as [calc_agostini_duplication\(\)](#). Where Agostini uses one empirical coefficient for every vehicle pair, Hofmans computes a pairwise coefficient directly from each pair's own audiences and duplication, so no coefficient has to be fitted to external data. Like Agostini's formula, it estimates total reach only, not the exposure distribution. It is unrelated to [calc_hofmans_accumulation\(\)](#), the same author's model for one vehicle with several insertions.

Usage

```
calc_hofmans_duplication(audiences, population, duplication_matrix)
```

Arguments

audiences	Numeric vector with the audience of each vehicle for one insertion, in people.
population	Population size, in people. No audience can exceed it; otherwise the function stops with an error.
duplication_matrix	Symmetric numeric matrix whose element $[i, j]$ is the number of people who are in the audience of both vehicle i and vehicle j (one insertion in each), in people (not a proportion). Diagonal values are ignored.

Details

For two vehicles, reach is exactly $A_1 + A_2 - A_{12}$. Equating that value to Agostini's formula gives the coefficient $k_{ij} = (A_i + A_j) / (A_i + A_j - A_{ij})$, which Hofmans applies to every pair of a plan with m vehicles:

$$R_m = \frac{(\sum_{i=1}^m A_i)^2}{\sum_{i=1}^m A_i + \sum_{i=1}^{m-1} \sum_{j=i+1}^m k_{ij} A_{ij}}.$$

The double sum runs over every pair of vehicles, each pair counted once.

The formula is empirical and can return a reach outside its logical range (below the largest audience, or above the population or the gross audience). This can happen because the duplications are not mutually consistent, but also because the formula is an approximation: five independent vehicles that each reach 50% of the population, with a duplication of 25% in every pair (data that come from a valid joint distribution, with a true reach of 96.875%), give a reach of 107.14%. The value is then returned unchanged, with a warning, so that an out-of-range result is not attributed automatically to the input data.

Value

A list of class "reach_hofmans_duplication" with components:

- reach: list with percent and people.
- gross_audience: sum of the vehicle audiences (duplicated people counted once per vehicle), in people.
- weighted_duplication: the sum $\sum_{i=1}^{m-1} \sum_{j=i+1}^m k_{ij} A_{ij}$, in people.
- n_vehicles: number of vehicles.

References

Aldás Manzano, J. (1998). Modelos de determinación de la cobertura y la distribución de contactos en la planificación de medios publicitarios impresos (Models for determining reach and exposure distribution in print media planning). Doctoral dissertation, Universidad de Valencia, Spain. Section 3.2.1.2, equations 3.59-3.62.

Hofmans, P. (1966). Measuring the cumulative net coverage of any combination of media. *Journal of Marketing Research*, 3(3), 269-278. doi:[10.1177/002224376600300307](https://doi.org/10.1177/002224376600300307)

Kim, H. G. (2005). A Canonical Sequential Aggregation Media Model. Doctoral dissertation, The University of Texas at Austin, pp. 44-45.

See Also

[calc_agostini_duplication\(\)](#), the single-coefficient formula this model refines pair by pair.

Examples

```
data(duplication_example)
do.call(calc_hofmans_duplication, duplication_example)
```

calc_mbd	<i>Multivariate Beta Binomial Distribution model</i>
----------	--

Description

Implements Cheong's (2007) Multivariate Beta Binomial Distribution (MBD). Vehicle co-exposure probabilities for pairs (observed) and for three or more vehicles (imputed from a Beta-Binomial fitted to the subset's own mean audience and mean pairwise duplication) are combined into the full 2^m joint zero/one exposure grid by Waring's (1792) inclusion-exclusion theorem. Vehicles are then peeled off one at a time, in reverse aggregation order: each remaining exposure pattern's row is expanded from a zero/one state into the peeled vehicle's own insertion-level Beta-Binomial distribution by a row-specific conditional allocation, and the two component distributions are convolved into a growing pseudo-vehicle.

Usage

```
calc_mbd(
  vehicles_data,
  duplications,
  aggregation_order = c("audience_desc", "given"),
  population = 1,
  tolerance = 1e-08
)
```

Arguments

vehicles_data	Data frame with columns insertions, R1 and R2, with the same convention as calc_csd() (R1 and R2 are proportions between 0 and 1, not percentages, R1 is strictly between zero and one, $R1 \leq R2 \leq 2 * R1 - R1^2$, and R2 may be NA only when insertions is one; the function stops with an explanatory error if R2 is outside that range). At most 12 vehicles are supported.
duplications	Symmetric matrix whose element $[i, j]$ is the proportion of the population that is exposed to both vehicle i and vehicle j (one insertion in each), a number between 0 and 1 (not a percentage). It is not the proportion exposed to each vehicle separately. The diagonal is ignored.
aggregation_order	Either "audience_desc" (vehicles in decreasing order of one-insertion reach R1, ties in the row order; duplications are not used to sort), "given" (the row order), or a permutation of the row indices. Vehicles are peeled off starting from the <i>last</i> position of this order, as in Cheong's worked examples. In MBD the order can change both the reach and the exposure distribution (see Details).
population	Number of people in the population (a count, not a proportion). It only converts probabilities into people: the people columns of the result are the probability times population, and with the default, 1, they equal the probability.
tolerance	Positive numerical tolerance for probability constraints. The Fréchet and triple-feasibility checks apply it relative to the smaller audience involved.

Details

`calc_mbd()` (Cheong, 2007) is unrelated to `fit_bbd_to_reach()`, which fits one Beta-Binomial to an externally given reach. MBD is the *Multivariate* Beta Binomial Distribution of several vehicles described here, whereas `fit_bbd_to_reach()` fits a single Beta-Binomial to a whole schedule.

Value

A `reach_mbd` object: a list with `reach` (probability, percent and people), `average_frequency`, the complete exposure distribution (contacts, probability, percent, people and `cumulative_probability`), the `aggregation_order` and `aggregation_rule`, the peeling steps and diagnostics, which report whether the final negative-probability safety net (Cheong's "MBD-ADJ") had to be engaged and how much probability mass it redistributed (`negative_mass_adjusted`, `cells_adjusted`).

What is, and is not, guaranteed

Cheong's dissertation reports that the inclusion-exclusion grid can produce small negative cell probabilities once four or more vehicles are combined, because the co-exposure probability of three or more vehicles is *imputed* (no closed-form multivariate distribution stands behind it) rather than observed. Cheong's four-vehicle example is fully corrected by a first-order consistency check (every combined-exposure probability is shrunk to remain below each of its immediate subsets); the five-vehicle example is not, and the dissertation states that resolving this in general would need increasingly complex higher-order checks that were never implemented, proposing iterative proportional fitting as future work. `calc_mbd()` therefore applies exactly the checks Cheong specifies (first order only) and, like Cheong's MBD-ADJ variant, sets any cell that is still negative to zero and redistributes that mass proportionally across the non-negative cells of the final collapsed distribution. This is reported in diagnostics; a non-zero value means the result for that schedule relies on this fallback rather than on a value Cheong verified as internally consistent without it. The function warns for four or more vehicles, the range where Cheong's own checks are known to be insufficient by themselves.

Cheong also reports that the aggregation order can change the collapsed distribution, that this was not investigated systematically, and that the model was evaluated on schedules of 12 vehicles because the computing time of the exposure grid becomes prohibitive above 13 vehicles. `calc_mbd()` stops above 12 vehicles for this reason.

Validation against the published example

For the three-vehicle example of Cheong (2007, p. 75), which Cheong, Leckenby and Eakin (2011, Table 4) also publish, the function reproduces the published distribution to within 0.002 in every cell, which is more than the rounding of the published values (0.0005). An independent reimplementing of the printed algorithm agrees with this function to $1e-4$, so the gap originates in the printed intermediate tables, which contain arithmetic inconsistencies: a product printed as $.024(.057) = .003$ (it is $.0014$), a sum of Beta-Binomial parameters of 7.294 (the printed formula gives 7.543), and an exclusive-grid sum of 1.001. The package tests fix the accepted deviation per cell. In the binomial limit of a vehicle's own distribution (R^2 equal to the reach under independence), the conditional distributions of the peeling step coincide, so the result of this model does not depend on the duplication between vehicles in that limit; this follows from Cheong's construction.

Of the eleven models Cheong evaluated (comScore 2003 data, 440 schedules), the three versions of MBD were the most accurate for reach alone (average percentage error in reach of 1.18% for the best version), but not for the complete exposure distribution, where `calc_canex()` (6.91% average percentage error) and the Conditional Beta Distribution model (`calc_cbd()`, 8.80%) were more accurate than the MBD versions (10.19% to 12.10%).

Domain of validity

In plain words. These models take the overlaps between vehicles (the duplications) from your data and use them to build the whole distribution of exposures. For that to be possible the overlaps you supply must be possible, and plausible together:

- Two vehicles cannot overlap more than the smaller of their audiences, nor less than their audiences force them to (two audiences of 60% of the population must overlap by at least 20%). If a duplication breaks these limits, the function stops with an error naming the pair: the data cannot be right and should be reviewed.
- The overlaps of three vehicles must also fit together. If they cannot, the function stops with an error naming the three vehicles.
- The models were built for the usual situation in which vehicles overlap at least as much as chance alone would produce, that is, at least the product of their audiences (two vehicles with audiences of 30% and 20% overlap by 6% if people chose vehicles independently). When a pair overlaps clearly less than that, the results can differ a lot from one model to another, and the function issues a warning that names the pair.

An error means that the inputs are impossible: correct them. A warning does not invalidate the calculation, but it asks for caution: check the duplication of the pair named, which is often a measurement problem (see below), and compare the result with another model. The rest of this section gives the technical detail.

The multivariate models (`calc_canex()`, `calc_csd()`, `calc_msad()`, `calc_cbd()` and `calc_mbd()`) take the observed one-insertion duplications as given and check them against the Fréchet bounds implied by the audiences and, for every triple of vehicles, against joint feasibility. The published methods have three limits that follow from this.

First, Hong (1998, p. 238) notes that the models he evaluates assume a duplication between vehicles not below the random one, the product of the audiences, and treats null observed duplications as measurement error. With a lower duplication the models can diverge considerably. The functions warn (`class mediaPlanR_low_duplication`) when the canonical correlation of a pair is below -0.1. That threshold is a decision of this package: the published reference plans (Kim 2005, Appendix B, and Hong 1998, Appendix E; 80 plans) include pairs slightly below the random duplication, with canonical correlations down to -0.071, and they raise no warning. For two vehicles with two insertions each, $R1 = 0.30$ and $R2 = 0.45$, a duplication of zero gives a reach of 0.90 under MSAD, 0.81 under MBD, 0.78 under CSD and CBD, and 0.79 under CANEX. Danaher's canonical expansion itself admits negative correlations. Hong (1998, pp. 238-239) treats a null duplication as a measurement error and replaces it by $\hat{p}_{ij} = K' p_i p_j$, where K' is the ratio between observed and random duplication over the pairs with non-zero observed duplication. The package does not implement this imputation: `calc_agostini_duplication()` and `calc_hofmans_duplication()` estimate reach from a duplication matrix and do not impute its elements. If duplications are imputed beforehand, the result must be checked against the bounds described below. A null duplication is not always a measurement error: it can also be a genuine observation.

Second, pairwise duplications that satisfy the Fréchet bounds can still be impossible jointly for three or more vehicles. For every triple of vehicles the functions check that the exposure to all three can have a non-negative probability, given the audiences and the three pairwise duplications, and stop with an error when it cannot. That condition is necessary but not sufficient for four or more vehicles, for which the published methods check feasibility at most to first order (Cheong 2007); `calc_mbd()` and `calc_cbd()` zero any negative cell that remains, and `calc_canex()` truncates it.

Third, a low duplication between vehicles with several insertions can make the target of a sequential model invalid. For two vehicles with two insertions each, $R1 = 0.5$ and $R2 = 0.75$, `calc_csd()` and `calc_cbd()` stop with a duplication of zero, and `calc_msad()` with a duplication of zero or 0.5, instead of altering an invalid target.

References

Cheong, Y. (2007). Multivariate Beta Binomial Distribution Model as a Web Media Exposure Model. Doctoral dissertation, The University of Texas at Austin.

Cheong, Y., Leckenby, J. D., & Eakin, T. (2011). Evaluating the multivariate beta binomial distribution for estimating magazine and Internet exposure frequency distributions. *Journal of Advertising*, 40(1), 7-23. <https://doi.org/10.2753/JOA0091-3367400101>

Waring, E. (1792). On the principles of translating algebraic quantities into probable relations and annuities. Cambridge. (The work Cheong, 2007, and Aldás Manzano, 1998, cite for the theorem; not to be confused with Waring's *Meditationes Algebraicae*, 1770.)

See Also

[calc_csd\(\)](#) and [calc_msad\(\)](#) for sequential aggregation models, and [calc_cbd\(\)](#) for the model that uses a canonical-expansion between-vehicle step instead of an imputed one.

Examples

```
# Cheong (2007), Chapter 4.2: three-vehicle conceptual example
data(mbd_cheong2007)
result <- do.call(calc_mbd, mbd_cheong2007)
result$reach
result$distribution
```

calc_metheringham

Reach and exposure distribution under the Metheringham model

Description

Implements Metheringham's (1964) model for a plan with several vehicles and several insertions per vehicle, in the version of Aldás Manzano (1998). Individuals are heterogeneous, with Beta-distributed exposure probabilities; vehicles are treated as homogeneous, so that duplication between different vehicles is equivalent to accumulation within one hypothetical "average" vehicle. The audience and the duplication of that average vehicle are the averages, over all pairs of insertions, of the observed audiences and duplications. They fix the two parameters of a Beta-Binomial distribution for the plan's $N = \text{sum}(\text{insertions})$ insertions.

Usage

```
calc_metheringham(audiences, insertions, duplication_matrix, population)
```

Arguments

audiences	Numeric vector with the audience of each vehicle, in people per insertion.
insertions	Non-negative integer vector with the number of insertions planned in each vehicle. The plan as a whole (all vehicles together) must have at least two insertions, for example two vehicles with one insertion each, or one vehicle with two insertions, because the model works with duplication between pairs of insertions.

duplication_matrix	Symmetric numeric matrix, in people. Element $[i, j]$ with $i \neq j$ is the number of people who are exposed to both one insertion in vehicle i and one insertion in vehicle j . The diagonal element $[i, i]$ is the audience duplicated between two insertions in the same vehicle i , that is, $2 * \text{audiences}[i]$ minus the audience accumulated by two insertions in vehicle i . Diagonal elements are needed only for vehicles with at least two insertions, and elements involving a vehicle with no insertions are ignored.
population	Population size, in people. No audience can exceed it; otherwise the function stops with an error.

Details

There are $N = \text{sum}(\text{insertions})$ insertions and choose($N, 2$) pairs of insertions: $\text{insertions}[i] * \text{insertions}[j]$ pairs between vehicles i and j , and choose($\text{insertions}[i], 2$) pairs within vehicle i . The model computes

1. the insertion-weighted mean audience $\bar{A}_1 = \sum_{i=1}^m n_i A_i / N$, with m the number of vehicles;
2. the pair-weighted mean duplication $\bar{D}$, over all pairs of insertions;
3. the mean audience accumulated by two insertions of the average vehicle, $\bar{A}_2 = 2\bar{A}_1 - \bar{D}$, which equals the average, over all pairs of insertions, of the audience reached by the pair;
4. alpha and beta by the Beta-Binomial estimators of `calc_beta_binomial()`, and the exposure distribution for N insertions.

The exposure distribution therefore ranges from zero to N exposures, and its mean equals the plan's gross number of exposures per person, $\sum_{i=1}^m n_i A_i / P$.

Value

A list of class "reach_metheringham" with components:

- reach, distribution, cumulative and parameters, as in `calc_beta_binomial()` evaluated for N insertions;
- mean_audience: insertion-weighted mean audience (A_1), in people;
- mean_duplication: pair-weighted mean duplication (D), in people;
- second_audience: mean audience accumulated by two insertions of the average vehicle (A_2), in people;
- opportunity_matrix: number of pairs of insertions between and within vehicles;
- opportunity_vector and duplication_vector: the upper triangles of the opportunity and duplication matrices, linearized row by row;
- total_insertions: the number of insertions N .

References

Aldás Manzano, J. (1998). Modelos de determinación de la cobertura y la distribución de contactos en la planificación de medios publicitarios impresos (Models for determining reach and exposure distribution in print media planning). Doctoral dissertation, Universidad de Valencia, Spain. Section 3.3.1.5 (several insertions per vehicle, pages 196-198) and Section 3.2.2.9 (one insertion per vehicle).

Metheringham, R. A. (1964). Measuring the net cumulative coverage of a print campaign. *Journal of Advertising Research*, 4(4), 23-28. doi:10.1080/00218499.1964.12519751

See Also

`calc_beta_binomial()`, called internally, for the univariate model; `calc_sainsbury()` and `calc_binomial()` for models that assume random duplication.

Examples

```
data(metheringham_example)
result <- do.call(calc_metheringham, metheringham_example)
result$reach$percent
result$parameters$alpha

# One insertion per vehicle reduces to Section 3.2.2.9 of Aldas Manzano
# (1998); the diagonal is not needed then
calc_metheringham(
  audiences = c(300000, 400000, 200000), insertions = c(1, 1, 1),
  duplication_matrix = matrix(c(NA, 150000, 90000,
                                150000, NA, 110000,
                                90000, 110000, NA), nrow = 3),
  population = 1000000
)$reach$percent
```

calc_msad

Morgensztern Sequential Aggregation Distribution model

Description

Implements the MSAD structure described by Leckenby and Rice (1986) and Kim (2005): each vehicle is expanded with a Beta-Binomial marginal, the reach of every sub-schedule is estimated with the Morgensztern formula, and the marginals are combined sequentially with a non-random, margin-preserving convolution whose zero cell equals one minus that reach. Kim specifies the MSAD algorithm but refers readers to Lee (1988) for its complete worked numerical example; Kim's own three-vehicle example is CSD, not MSAD.

Usage

```
calc_msad(
  vehicles_data,
  duplications,
  aggregation_order = c("audience_desc", "given"),
  population = 1,
  tolerance = 1e-10
)
```

Arguments

vehicles_data Data frame with one row per vehicle (at least two rows) and columns `insertions` (planned insertions, a positive integer), `R1` and `R2`. `R1` is the reach after one insertion, a proportion between 0 and 1 (not a percentage) and strictly between zero and one, and `R2` the cumulative reach after two insertions (the proportion exposed at least once in two insertions), a proportion with $R1 \leq R2 \leq 2 * R1 - R1^2$. If `R2` is outside that range the function stops with an explanatory error. `R2` may be NA only when `insertions` is one.

duplications	Symmetric matrix whose element $[i, j]$ is the proportion of the population that is exposed to both vehicle i and vehicle j (one insertion in each), a number between 0 and 1 (not a percentage). It is not the proportion exposed to each vehicle separately. The diagonal is ignored.
aggregation_order	Either "audience_desc" (Kim's larger-audience- first rule: vehicles in decreasing order of one-insertion reach R_1), "given" (the row order), or a permutation of the row indices. The vehicles are combined one at a time in this order. The order does not change the reach of the whole schedule but can change the exposure distribution.
population	Number of people in the population (a count, not a proportion). It only converts probabilities into people: the people columns of the result are the probability times population, and with the default, 1, they equal the probability.
tolerance	Positive numerical tolerance for probability constraints. The Fréchet and triple-feasibility checks apply it relative to the smaller audience involved.

Details

For a subset of m vehicles, the Morgensztern reach is

$$R_m = \frac{(\sum_{i=1}^m R_{n_i})^2}{\sum_{i=1}^m R_{n_i} + \sum_{i=1}^{m-1} \sum_{j=i+1}^m K_{ij} A_{ij} R_{n_i} R_{n_j} / (A_i A_j)},$$

with $K_{ij} = (A_i + A_j) / (A_i + A_j - A_{ij})$, where R_{n_i} is the reach of vehicle i for its own n_i insertions, A_i its one-insertion audience and A_{ij} the one-insertion duplication of vehicles i and j , all as proportions of the population. The double sum runs over every pair of vehicles. At every aggregation step the joint table is conformed to the two input marginal distributions and to the Morgensztern union reach. Consequently all probabilities remain non-negative, the margins are preserved, and the zero-exposure probability is exactly $1 - R_m$.

The function reproduces the three-vehicle numerical example of Lee (1988, pp. 81-93; two insertions per vehicle, aggregation in the order Lee obtains with his total-duplication criterion, aggregation_order = "given" with the vehicles listed in that order) to within 0.0011 in every cell of the exposure distribution, which Lee prints to three decimals.

MSAD is intended to reduce, but does not guarantee the elimination of, the declining reach phenomenon (Leckenby & Rice, 1986): the reach that some exposure-distribution models estimate for a schedule can fall when insertions or vehicles are added, which cannot happen in reality. The reach formula can also be incompatible with the supplied marginal distributions. In that case the function stops and reports the feasible interval instead of silently altering the target. The aggregation order can change the exposure distribution even when the final schedule reach is unchanged.

Value

A reach_msad object: a list with reach (probability, percent and people), average_frequency, the complete exposure distribution (contacts, probability, percent, people and cumulative_probability), the vehicle_marginals, the one-insertion vehicle_reach, the aggregation_order and aggregation_rule, the aggregation steps and numerical diagnostics (see [calc_csd\(\)](#) for gross_mean_contacts, distribution_mean_contacts and mean_error).

Domain of validity

In plain words. These models take the overlaps between vehicles (the duplications) from your data and use them to build the whole distribution of exposures. For that to be possible the overlaps you supply must be possible, and plausible together:

- Two vehicles cannot overlap more than the smaller of their audiences, nor less than their audiences force them to (two audiences of 60% of the population must overlap by at least 20%). If a duplication breaks these limits, the function stops with an error naming the pair: the data cannot be right and should be reviewed.
- The overlaps of three vehicles must also fit together. If they cannot, the function stops with an error naming the three vehicles.
- The models were built for the usual situation in which vehicles overlap at least as much as chance alone would produce, that is, at least the product of their audiences (two vehicles with audiences of 30% and 20% overlap by 6% if people chose vehicles independently). When a pair overlaps clearly less than that, the results can differ a lot from one model to another, and the function issues a warning that names the pair.

An error means that the inputs are impossible: correct them. A warning does not invalidate the calculation, but it asks for caution: check the duplication of the pair named, which is often a measurement problem (see below), and compare the result with another model. The rest of this section gives the technical detail.

The multivariate models (`calc_canex()`, `calc_csd()`, `calc_msad()`, `calc_cbd()` and `calc_mbd()`) take the observed one-insertion duplications as given and check them against the Fréchet bounds implied by the audiences and, for every triple of vehicles, against joint feasibility. The published methods have three limits that follow from this.

First, Hong (1998, p. 238) notes that the models he evaluates assume a duplication between vehicles not below the random one, the product of the audiences, and treats null observed duplications as measurement error. With a lower duplication the models can diverge considerably. The functions warn (class `mediaPlanR_low_duplication`) when the canonical correlation of a pair is below -0.1. That threshold is a decision of this package: the published reference plans (Kim 2005, Appendix B, and Hong 1998, Appendix E; 80 plans) include pairs slightly below the random duplication, with canonical correlations down to -0.071, and they raise no warning. For two vehicles with two insertions each, $R1 = 0.30$ and $R2 = 0.45$, a duplication of zero gives a reach of 0.90 under MSAD, 0.81 under MBD, 0.78 under CSD and CBD, and 0.79 under CANEX. Danaher's canonical expansion itself admits negative correlations. Hong (1998, pp. 238-239) treats a null duplication as a measurement error and replaces it by $\hat{p}_{ij} = K' p_i p_j$, where K' is the ratio between observed and random duplication over the pairs with non-zero observed duplication. The package does not implement this imputation: `calc_agostini_duplication()` and `calc_hofmans_duplication()` estimate reach from a duplication matrix and do not impute its elements. If duplications are imputed beforehand, the result must be checked against the bounds described below. A null duplication is not always a measurement error: it can also be a genuine observation.

Second, pairwise duplications that satisfy the Fréchet bounds can still be impossible jointly for three or more vehicles. For every triple of vehicles the functions check that the exposure to all three can have a non-negative probability, given the audiences and the three pairwise duplications, and stop with an error when it cannot. That condition is necessary but not sufficient for four or more vehicles, for which the published methods check feasibility at most to first order (Cheong 2007); `calc_mbd()` and `calc_cbd()` zero any negative cell that remains, and `calc_canex()` truncates it.

Third, a low duplication between vehicles with several insertions can make the target of a sequential model invalid. For two vehicles with two insertions each, $R1 = 0.5$ and $R2 = 0.75$, `calc_csd()` and `calc_cbd()` stop with a duplication of zero, and `calc_msad()` with a duplication of zero or 0.5, instead of altering an invalid target.

References

Leckenby, J. D., & Rice, M. D. (1986). The declining reach phenomenon in exposure distribution models. *Journal of Advertising*, 15(3), 13-20. doi:10.1080/00913367.1986.10673014

Kim, H. G. (2005). A Canonical Sequential Aggregation Media Model. Doctoral dissertation, The University of Texas at Austin, pp. 65-71. Kim identifies the detailed MSAD numerical example as Lee, H.-K. (1988), Sequential aggregation advertising media models, unpublished doctoral dissertation, The University of Texas at Austin, pp. 81-101; the numerical example itself occupies pp. 81-93.

Aldás Manzano, J. (1998). Modelos de determinación de la cobertura y la distribución de contactos en la planificación de medios publicitarios impresos (Models for determining reach and exposure distribution in print media planning). Doctoral dissertation, Universidad de Valencia, Spain. Section 3.3.2.2, equation 3.106, for the Morgensztern reach formula.

See Also

`calc_csd()` for the canonical alternative and `calc_mbd()` and `calc_cbd()` for models that peel vehicles into insertion-level distributions.

Examples

```
# MSAD calculation derived from the inputs of Kim's CSD example. The
# resulting MSAD values are not presented by Kim as a published benchmark.
data(msad_kim2005)
result <- do.call(calc_msad, msad_kim2005)
result$reach
```

calc_sainsbury

Reach and exposure distribution under the Sainsbury model

Description

Implements the Sainsbury model, which Aldás Manzano (1998, Section 3.2.2.2) reports as developed at the London Press Exchange and formalized by Caffyn and Sagovsky (1963), to calculate reach and the exposure distribution (and its cumulative counterpart) of a set of advertising vehicles, with one or more insertions in each of them (argument `insertions`). The model assumes random duplication *and* random accumulation, homogeneous individual exposure probabilities and heterogeneous vehicle exposure probabilities. The probability that an individual is exposed to one insertion in vehicle *i* is therefore the ratio between the vehicle's audience and the population. Because duplication and accumulation are both random, every insertion – whether in a different vehicle or a repeat insertion in the same one – is an independent Bernoulli trial with that vehicle's own exposure probability, and the number of exposures follows a Poisson-binomial distribution (the distribution of the number of successes in independent trials whose success probabilities differ).

Usage

```
calc_sainsbury(audiences, population, insertions = rep(1L, length(audiences)))
```

Arguments

<code>audiences</code>	Numeric vector with the audience of each vehicle, in people per insertion.
<code>population</code>	Population size, in people. No audience can exceed it; otherwise the function stops with an error.

insertions Non-negative integer vector with the number of insertions planned in each vehicle. It defaults to one insertion per vehicle, the model's original scope; values above one apply the same independence hypothesis to repeat insertions in the same vehicle.

Details

The exposure distribution is computed exactly, by combining one insertion at a time (a dynamic convolution of the per-insertion Bernoulli distributions). The number of calculations grows with the square of the number of insertions N , so doubling the insertions roughly quadruples the work, which is negligible for any realistic plan. Reach equals $1 - \prod_{i=1}^m (1 - A_i/P)^{n_i}$, where m is the number of vehicles, A_i is the audience of vehicle i , n_i its number of insertions and P the population.

Value

A list of class "reach_sainsbury" with components:

- **reach**: list with percent and people.
- **distribution**: list with percent and people, one value for each number of exposures from one to the total number of insertions.
- **cumulative**: list with percent and people, for individuals exposed at least once, at least twice, and so on.

References

Aldás Manzano, J. (1998). Modelos de determinación de la cobertura y la distribución de contactos en la planificación de medios publicitarios impresos (Models for determining reach and exposure distribution in print media planning). Doctoral dissertation, Universidad de Valencia, Spain. Section 3.2.2.2 for one insertion per vehicle; Section 3.3.1.2, equations 3.86-3.87, for several insertions (the multivariable binomial model with independent vehicles that Aldás Manzano attributes to Chandon).

Chandon, J.-L. (1976). A comparative study of media exposure models. Unpublished doctoral dissertation, Northwestern University, Evanston, IL. (Aldás Manzano, 1998, cites Chandon's work as Chandon, 1985.)

See Also

[calc_binomial\(\)](#) for the homogeneous-vehicle counterpart, [calc_beta_binomial\(\)](#) and [calc_metherringham\(\)](#) for models with heterogeneous individuals, and [estimate_reach\(\)](#) to run this model from a [media_plan\(\)](#) object.

Examples

```
audiences <- c(300000, 400000, 200000)
population <- 1000000
result <- calc_sainsbury(audiences, population)
result$reach$percent
result$distribution$people

# The same three vehicles with several insertions each
calc_sainsbury(audiences, population, insertions = c(4, 6, 10))
```

calibrate_bbd

*Calibrate a Beta-Binomial effective-reach model***Description**

Calibrates a Beta-Binomial exposure distribution so that it reproduces a target effective reach. The reach after one insertion fixes the mean of the Beta distribution, and only its concentration is calibrated, by a one-dimensional search instead of a two-dimensional grid over both shape parameters.

Usage

```
calibrate_bbd(
  R1,
  target_reach,
  frequency,
  max_insertions,
  type = c("at_least", "exact"),
  tolerance = 1e-06
)
```

Arguments

R1	Reach after one insertion (the vehicle's audience), as a proportion (not a percentage) strictly between zero and one.
target_reach	Target proportion of the population (not a percentage), strictly between zero and one. With type = "at_least" it is the effective reach: the proportion exposed frequency times or more. With type = "exact" it is the proportion exposed exactly frequency times (which is then not a reach in the usual sense).
frequency	Positive integer: the number of exposures that defines the target (frequency or more exposures if type = "at_least", exactly frequency exposures if type = "exact").
max_insertions	Maximum number of insertions considered, an integer of at least frequency.
type	What target_reach refers to. "at_least" (default): the probability of frequency or more exposures (for frequency = 2, being exposed twice, three times, and so on; this is the effective reach). "exact": the probability of exactly frequency exposures (for frequency = 2, exactly twice).
tolerance	Desired absolute probability error; it only determines the converged flag.

Details

For every number of insertions n from frequency to max_insertions, the function searches for the concentration of the Beta-Binomial, with mean R1, whose probability equals target_reach, first on a grid of the log concentration and then by `stats::optimize()` around the best grid point. It returns the number of insertions with the smallest error, the smaller n in case of ties.

Value

A bbd_calibration object: a list with alpha, beta, concentration, insertions (the selected n), R1, frequency, type, target_reach, predicted_reach, error, converged, the full distribution (contacts, probability and cumulative_probability) and the candidates evaluated for each n .

See Also

[calc_beta_binomial\(\)](#) for the model itself and [fit_bbd_to_reach\(\)](#) to fit a Beta-Binomial to a whole schedule's reach.

Examples

```
# A first insertion reaches 30% and the share exposed two or more times
# should be 20%
calibrate_bbd(R1 = 0.30, target_reach = 0.20, frequency = 2,
              max_insertions = 6)
```

canex_example

Illustrative inputs for the CANEX model

Description

List of arguments ready for `do.call()` with [calc_canex\(\)](#): two vehicles with three and two insertions. Original illustrative data, not derived from any published source. For inputs taken from a published worked example, see [csd_kim2005](#).

Usage

```
canex_example
```

Format

A list with the components:

vehicles_data Data frame with columns `insertions`, `R1` and `R2` (reach after one and two insertions, as proportions between 0 and 1).

duplications Symmetric matrix of proportions of the population exposed to both vehicles of each pair (one insertion in each).

population Population size, in people.

Examples

```
data(canex_example)
do.call(calc_canex, canex_example)
```

compare_reach_models	<i>Compare reach estimates under several models</i>
----------------------	---

Description

Compare reach estimates under several models

Usage

```
compare_reach_models(plan, models = c("sainsbury", "binomial"))
```

Arguments

plan	A media_plan object.
models	Character vector containing "sainsbury" and/or "binomial"; see estimate_reach() .

Value

A data frame with one row per model and columns model, reach_probability, reach_percent, reach_people and average_frequency.

Examples

```
plan <- media_plan(
  data.frame(channel = c("TV", "Radio"), audience = c(300000, 180000),
             insertions = c(4, 6), cost_per_insertion = c(18000, 3500)),
  population = 1000000
)
compare_reach_models(plan)
```

csd_example	<i>Illustrative inputs for the CSD model</i>
-------------	--

Description

List of arguments ready for `do.call(calc_csd, csd_example)`: a small three-vehicle scenario. Original illustrative data, not derived from any published source. For a dataset that reproduces published inputs for literature validation, see [csd_kim2005](#).

Usage

```
csd_example
```

Format

A list with the components:

vehicles_data Three rows with columns insertions, R1 and R2.

duplications Symmetric matrix of proportions of the population exposed to both vehicles of each pair (one insertion in each).

aggregation_order The order of aggregation, 1:3.

See Also

[calc_csd\(\)](#), [csd_kim2005](#), [msad_example](#)

Examples

```
data(csd_example)
result <- do.call(calc_csd, csd_example)
result$distribution
```

csd_kim2005

Published inputs for the CSD model (Kim, 2005)

Description

Minimal factual inputs (insertion counts, reach and pairwise duplication figures) reproduced from Kim (2005), Tables 4.2.2.1-4.2.2.10, for the three-vehicle Canonical Sequential Aggregation example with the aggregation order of the worked example (labeled "TD forward" by Kim). They are included solely so that users can verify that [calc_csd\(\)](#) reproduces the published result; exact calculations keep more precision than the intermediate values rounded in the dissertation. They are bare numeric values reused for validation, not a reproduction of the dissertation. The audiences and reaches originate from comScore Media Metrix data of September 2003, which Kim (2005) uses and Cheong, Leckenby and Eakin (2011, Table 5) also publish.

Usage

```
csd_kim2005
```

Format

A list ready for `do.call(calc_csd, csd_kim2005)` with the components:

vehicles_data Three rows with columns insertions, R1 and R2.

duplications Symmetric matrix of proportions of the population exposed to both vehicles of each pair (one insertion in each).

aggregation_order The published forward order, 1:3.

References

Kim, H. G. (2005). A Canonical Sequential Aggregation Media Model. Doctoral dissertation, The University of Texas at Austin, pp. 80-97.

See Also

[calc_csd\(\)](#), [csd_example](#), [msad_kim2005](#)

Examples

```
data(csd_kim2005)
result <- do.call(calc_csd, csd_kim2005)
result$distribution
```

duplication_example	<i>Illustrative inputs for the Agostini and Hofmans duplication models</i>
---------------------	--

Description

List of arguments ready for `do.call()` with `calc_agostini_duplication()` and `calc_hofmans_duplication()`, the two ad hoc duplication models for several vehicles with one insertion each. Original illustrative data, not derived from any published source.

Usage

```
duplication_example
```

Format

A list with the components:

audiences Numeric vector with the audience of each of three vehicles, in people per insertion.

population Population size, in people.

duplication_matrix Symmetric matrix whose element $[i, j]$ is the number of people in the audience of both vehicle i and vehicle j , in people. The diagonal is not used.

See Also

`calc_agostini_duplication()`, `calc_hofmans_duplication()`, `hofmans_accumulation_example`

Examples

```
data(duplication_example)
do.call(calc_agostini_duplication, duplication_example)
do.call(calc_hofmans_duplication, duplication_example)
```

estimate_reach	<i>Estimate reach and exposure distribution for a media plan</i>
----------------	--

Description

A `media_plan` front end for `calc_sainsbury()` and `calc_binomial()`: it reads audience, insertions and population from the plan and calls the requested model directly, so both share a single implementation.

Usage

```
estimate_reach(plan, model = c("sainsbury", "binomial"))
```

Arguments

plan	A media_plan object.
model	"sainsbury" (default) runs calc_sainsbury() : heterogeneous vehicle probabilities combined by the exact Poisson-binomial convolution. "binomial" runs calc_binomial() : every insertion has the plan's insertion-weighted mean probability. Both assume random duplication <i>and</i> random accumulation (a repeat insertion in the same vehicle is as independent as an insertion in a different vehicle); see vignette("mediaPlanR-intro") .

Details

Why only these two models: a media_plan holds, for each channel, only an audience, a number of insertions and a cost. [calc_sainsbury\(\)](#) and [calc_binomial\(\)](#) are the models that need nothing else. The other models need data that a plan does not contain: the observed duplication between vehicles ([calc_agostini_duplication\(\)](#), [calc_hofmans_duplication\(\)](#), [calc_metheringham\(\)](#), [calc_canex\(\)](#), [calc_csd\(\)](#), [calc_msad\(\)](#), [calc_cbd\(\)](#), [calc_mbd\(\)](#)) or the reach after two insertions ([calc_beta_binomial\(\)](#), [calc_hofmans_accumulation\(\)](#)). Run those models directly with their own inputs. Comparing the two models here shows the effect of unequal vehicle audiences at constant random duplication (see [compare_reach_models\(\)](#)); comparing either with observed data, or with another model, is done with [evaluate_exposure_model\(\)](#).

Value

A media_reach object: a list with the model, the population, reach (a data frame with probability, percent and people), distribution (contacts, probability, percent and people for zero up to the total number of insertions N), cumulative (min_contacts, probability, percent and people for at least min_contacts exposures), average_frequency (average exposures among the people reached) and model parameters.

See Also

[calc_sainsbury\(\)](#) and [calc_binomial\(\)](#), called directly by this function. For a Negative-Binomial count approximation, call [nbd_exposure_distribution\(\)](#) with your own mean_contacts: it is scoped to continuous exposure processes, not to finite insertion schedules, so it is not offered here or in [optimize_media_plan\(\)](#).

Examples

```
plan <- media_plan(
  data.frame(channel = c("TV", "Radio"), audience = c(300000, 180000),
             insertions = c(4, 6), cost_per_insertion = c(18000, 3500)),
  population = 1000000
)
estimate_reach(plan)
estimate_reach(plan, model = "binomial")$distribution
```

```
evaluate_exposure_model
```

Evaluate an observed exposure distribution against a model prediction

Description

Computes Kim's relative reach error and exposure-distribution error for one or more schedules, together with model-agnostic diagnostics. Observed data are always supplied by the analyst; the function never treats model inputs or fitted values as observations.

Usage

```
evaluate_exposure_model(
  observed,
  predicted,
  observed_scale,
  predicted_scale = c("auto", "count", "probability", "percent"),
  schedule_col = NULL,
  sum_tolerance = 0.005
)
```

Arguments

observed	Data frame with columns contacts and observed. It must include an explicit zero-exposure row and every integer exposure level up to its maximum. For several schedules, also include the column named by schedule_col.
predicted	Either a supported model result whose distribution is a data frame with contacts and probability or percent columns (for example the results of <code>calc_csd()</code> , <code>calc_msad()</code> , <code>calc_cbd()</code> , <code>calc_mbd()</code> , <code>calc_canex()</code> and <code>estimate_reach()</code>), or a data frame with columns contacts and predicted. Data-frame predictions must use the same schedule column when schedule_col is supplied.
observed_scale	Required declaration of the observed column: "count", "probability", or "percent". Counts may be weighted and are normalized within schedule.
predicted_scale	Scale of a predicted data frame. The default "auto" is allowed only for supported model objects, whose scale is known.
schedule_col	Optional name of a schedule identifier column present in both data frames. Model objects represent one schedule and therefore require schedule_col = NULL.
sum_tolerance	Maximum absolute departure from probability mass one accepted for probability/percent inputs. The default accepts ordinary publication rounding but rejects incomplete distributions.

Details

For schedule i , the function calculates

$$e_{R,i} = |R_i^{obs} - R_i^{pred}| / R_i^{obs}$$

and

$$e_{P,i} = \sum_{j \geq 1} |p_{ij}^{obs} - p_{ij}^{pred}| / R_i^{obs}.$$

where R_i is the reach (one minus the share with zero exposures) and p_{ij} the share exposed exactly j times. `kim_aer` (Kim's average percentage error in reach, AER) and `kim_ape` (Kim's average percentage error in the exposure distribution, APE) are the means of $e_{R,i}$ and $e_{P,i}$ across schedules, reported here as proportions (multiply by 100 for percentages). They are descriptive predictive-error measures, not inferential tests.

The other elements of summary are, averaged over schedules: `mean_total_variation`, half the sum over all exposure levels (including zero) of the absolute difference between observed and predicted probabilities; `mean_cell_mae`, the mean absolute difference per exposure level; `mean_reach_absolute_error`, $|R_i^{obs} - R_i^{pred}|$; and `mean_contact_bias`, predicted minus observed mean number of exposures per person (positive when the model overestimates exposures).

Exact support equality is required. Open-tail NBD output is rejected because a cell such as 10+ is not equivalent to an exact ten-exposure cell. To evaluate a censored distribution, the analyst must first collapse observed and predicted tails identically and provide explicit data frames.

Value

An `exposure_model_evaluation` object with `summary`, `by_schedule`, the aligned probability table, and input diagnostics.

References

Kim, H. G. (2005). A Canonical Sequential Aggregation Media Model. Doctoral dissertation, The University of Texas at Austin, pp. 117-118.

Examples

```
# Observations must be supplied explicitly. These are percentages from
# Kim's Table 4.2.2.10.
observed <- data.frame(
  contacts = 0:6,
  observed = c(38.41, 17.89, 39.66, 2.67, 1.36, 0, 0)
)
predicted <- data.frame(
  contacts = 0:6,
  predicted = c(38.20, 18.57, 39.18, 2.49, 1.51, 0.05, 0.01)
)
evaluation <- evaluate_exposure_model(
  observed, predicted,
  observed_scale = "percent", predicted_scale = "percent"
)
evaluation$summary[c("kim_aer", "kim_ape")]

# A fitted model object can be passed directly as the prediction.
data(csd_kim2005)
csd <- do.call(calc_csd, csd_kim2005)
evaluate_exposure_model(observed, csd, observed_scale = "percent")
```

fit_bbd_to_reach

Fit a Beta-Binomial distribution to an external reach estimate

Description

Fits one Beta-Binomial exposure distribution to a whole schedule so that it preserves the insertion-weighted mean exposure probability and reproduces a schedule reach supplied by the analyst. The reach can come from an ad hoc formula such as `calc_hofmans_duplication()` or the Morgensztern formula, or from any other independently justified estimator. This is the "estimation by mean and zeros" step of the Hofmans Beta-Binomial procedure; it is not the Morgensztern Sequential Aggregation Distribution (`calc_msad()`).

Usage

```
fit_bbd_to_reach(
  insertions,
  audiences,
  reach,
  population,
  precision = 100,
  max_iter = 100
)
```

Arguments

insertions	Numeric vector with the number of insertions of each vehicle (positive integers).
audiences	Numeric vector with the audience of each vehicle, in people per insertion.
reach	External schedule reach, in people.
population	Population size, in people. No audience can exceed it; otherwise the function stops with an error.
precision	Convergence criterion, in people. The default is 100.
max_iter	Maximum number of iterations of the root finder.

Details

When it is useful: several models (`calc_agostini_duplication()`, `calc_hofmans_duplication()`, the Morgensztern formula) and many real sources (a panel, a syndicated survey, planning software) give only the reach of a schedule, not how many times people are exposed. This function completes the picture: it finds the Beta-Binomial exposure distribution with that reach and the mean number of exposures implied by the audiences, so that the frequency distribution, the average frequency and the effective reach can be read from it. It is a bridge from a reach-only figure to a distribution, not a model of duplication: the result is only as good as the reach supplied and as the assumption that the Beta-Binomial shape suits the schedule.

With n_i insertions of audience A_i in vehicle i , the insertion-weighted mean exposure probability is $\bar{p} = \sum_{i=1}^m n_i A_i / (P \sum_{i=1}^m n_i)$, where m is the number of vehicles and P is the population (with the audiences A_i in people). The Beta-Binomial has $N = \sum_{i=1}^m n_i$ trials and shape parameters $\alpha = c\bar{p}$ and $\beta = c(1 - \bar{p})$, so its mean is preserved for every concentration $c > 0$. The function solves, with `stats::uniroot()` on the log scale of c , for the concentration whose reach, $1 - \Pr(K = 0)$, equals the external reach.

The reach a Beta-Binomial with this mean can take lies between the polarized limit ($c \rightarrow 0$), $\bar{p}P$, and the binomial limit ($c \rightarrow \infty$), $(1 - (1 - \bar{p})^N)P$. An external reach outside that interval, beyond precision, is rejected. At either limit the distribution is computed directly and alpha and beta are 0 or Inf.

Aldás Manzano (1998) starts the original iterative procedure from an arbitrary initial value of alpha; root finding needs no starting value.

Value

A list of class "bbd_reach_fit" with components:

- parameters: list with alpha, beta, N (total insertions), m (number of vehicles), population, iterations, converged, fit_type("beta_binomial", "polarized_limit" or "binomial_limit"), mean_probability and feasible_reach (the admissible interval, in people).
- reach: list with the external reach, the fitted Beta-Binomial reach and their difference, in people.
- distribution: data frame with contacts (0 to N), probability and cumulative_probability.
- iteration_history: data frame with the evaluations of the root finder.

References

Aldás Manzano, J. (1998). Modelos de determinación de la cobertura y la distribución de contactos en la planificación de medios publicitarios impresos (Models for determining reach and exposure distribution in print media planning). Doctoral dissertation, Universidad de Valencia, Spain. Section 3.3.1.10 (the Hofmans Beta-Binomial method of Leckenby and Boyd, pages 208-209) and Section 3.3.2.2 (the Morgensztern reach formula).

See Also

`calc_hofmans_duplication()` and `calc_agostini_duplication()`, whose reach this function can complete with a distribution. For models that estimate the exposure distribution without an external reach, see `calc_sainsbury()` and `calc_binomial()` (they need only audiences, population and insertions), `calc_beta_binomial()` (also needs the reach after two insertions) and `calc_metheringham()` (also needs the observed duplications).

Examples

```
data(bbd_reach_example)
fit <- do.call(fit_bbd_to_reach, bbd_reach_example)
fit

# From a reach-only formula to an exposure distribution: the reach of
# Hofmans' duplication model, completed with a Beta-Binomial distribution
data(duplication_example)
hofmans <- do.call(calc_hofmans_duplication, duplication_example)
fit_bbd_to_reach(
  insertions = c(1, 1, 1), audiences = duplication_example$audiences,
  reach = hofmans$reach$people, population = duplication_example$population
)
```

fit_nbd_exposure	<i>Fit a Negative-Binomial exposure-count model</i>
------------------	---

Description

Estimates the mean and Gamma heterogeneity parameter from observed person- level exposure counts by maximum likelihood. When the data are not overdispersed relative to Poisson, the fit returns the Poisson boundary.

Usage

```
fit_nbd_exposure(counts, report_max = NULL, conf_level = 0.95)
```

Arguments

counts	Vector of observed non-negative integer exposure counts.
report_max	Optional open-tail threshold passed to nbd_exposure_distribution() .
conf_level	Confidence level of the Wald interval for size, computed on the log scale of size, when an interior Negative-Binomial solution exists.

Details

It is meant for data in which each person has a count of exposures that is not limited by a number of insertions, such as page views or ad-server impressions per person in a period. From those counts it estimates the mean and the heterogeneity between people, and from them the reach and the frequency distribution, which can be extended to scenarios with [nbd_exposure_distribution\(\)](#). It is not used for plans with a fixed number of insertions.

Value

An `nbd_exposure_fit` object containing estimates, likelihood diagnostics, observed frequencies, and the fitted distribution.

Examples

```
counts <- c(rep(0, 40), rep(1, 25), rep(2, 15), rep(3, 8), 5, 7)
fit <- fit_nbd_exposure(counts)
fit$estimates
```

hofmans_accumulation_example	<i>Illustrative inputs for the Hofmans accumulation model</i>
------------------------------	---

Description

List of arguments ready for `do.call()` with [calc_hofmans_accumulation\(\)](#). One vehicle with several insertions: the "accumulation" domain. For the other Hofmans model (several vehicles, one insertion each), see [duplication_example](#). Original illustrative data, not derived from any published source.

Usage

```
hofmans_accumulation_example
```

Format

A list with the components:

R1 Reach after the first insertion, as a proportion.

R2 Cumulative reach after the second insertion, as a proportion.

insertions Number of insertions up to which the cumulative reach is calculated.

See Also

[calc_hofmans_accumulation\(\)](#), [duplication_example](#)

Examples

```
data(hofmans_accumulation_example)
do.call(calc_hofmans_accumulation, hofmans_accumulation_example)
```

mbd_cheong2007

Published inputs for the MBD model (Cheong, 2007)

Description

Minimal factual inputs (insertion counts, reach and pairwise duplication figures) reproduced from Cheong (2007), Chapter 4.2, for the three-vehicle conceptual example: vehicle A (2 insertions), vehicle B (1 insertion) and vehicle C (3 insertions). They are included solely so that users can verify that [calc_mbd\(\)](#) reproduces the published result up to the differences described in [calc_mbd\(\)](#) (the printed intermediate tables contain arithmetic inconsistencies, so the published distribution is matched only within 0.002 per cell). This is the example of Cheong's dissertation that is fully specified and that the implementation computes without the negative-probability safety net. They are bare numeric values reused for validation, not a reproduction of the dissertation. The figures originate from Simmons Market Research Bureau (SMRB) magazine audience data of 1979; the same example appears in Kim (1994, pp. 125-139) and in Cheong, Leckenby and Eakin (2011, Tables 2 to 4).

Usage

```
mbd_cheong2007
```

Format

A list ready for `do.call(calc_mbd, mbd_cheong2007)` with the components:

vehicles_data Three rows with columns `insertions`, `R1` and `R2` (`R2` is NA for the single-insertion vehicle).

duplications Symmetric matrix of proportions of the population exposed to both vehicles of each pair (one insertion in each).

aggregation_order 1:3, Cheong's own worked order.

References

Cheong, Y. (2007). Multivariate Beta Binomial Distribution Model as a Web Media Exposure Model. Doctoral dissertation, The University of Texas at Austin, Ch. 4.2.

Cheong, Y., Leckenby, J. D., & Eakin, T. (2011). Evaluating the multivariate beta binomial distribution for estimating magazine and Internet exposure frequency distributions. *Journal of Advertising*, 40(1), 7-23. <https://doi.org/10.2753/JOA0091-3367400101>

See Also

[calc_mbd\(\)](#), [mbd_example](#)

Examples

```
data(mbd_cheong2007)
result <- do.call(calc_mbd, mbd_cheong2007)
result$distribution
```

mbd_example

Illustrative inputs for the MBD model

Description

List of arguments ready for `do.call(calc_mbd, mbd_example)`: a small two-vehicle scenario. Original illustrative data, not derived from any published source. Two vehicles avoid the Beta-Binomial imputation of co-exposure that three or more vehicles require. For a three-vehicle literature-validation benchmark, see [mbd_cheong2007](#).

Usage

```
mbd_example
```

Format

A list with the components:

vehicles_data Two rows with columns insertions, R1 and R2.

duplications Symmetric matrix of proportions of the population exposed to both vehicles of each pair (one insertion in each).

aggregation_order The order of aggregation, 1:2.

See Also

[calc_mbd\(\)](#), [mbd_cheong2007](#)

Examples

```
data(mbd_example)
result <- do.call(calc_mbd, mbd_example)
result$distribution
```

Description

Reproducible tools for cross-media reach, contact-frequency distributions, target-audience metrics and budget-constrained reach optimization. The package combines a validated media-plan data contract with the classical reach and exposure-distribution models of the media-planning literature.

Scope of the validation

The models can be applied to any mix of media, but the published evidence on their accuracy comes mainly from magazine and Internet data. Their accuracy for a particular combination of media and population should be checked against observed data, for example with `evaluate_exposure_model()`.

Notation and units

The same quantity has the same meaning in every function. Names follow the sources of each model only where the literature fixes them.

- *Population* (number of people in the planning universe): population everywhere. In `audience_metrics()`, `gross_universe` is the whole population and `target_universe` the people who belong to the target.
- *Audience* (people exposed by one insertion of a vehicle): audience in `media_plan()` and audiences (one value per vehicle) in the `calc_*()` functions, always in people. As a proportion of the population it is R1.
- *Reach after the first and second insertion of the same vehicle*: R1 and R2 (Kim, 2005, and the sources of each model), always as proportions of the population, in `calc_beta_binomial()`, `calc_hofmans_accumulation()`, `calibrate_bbd()`, `calc_canex()` and the sequential models. R2 is cumulative: the people exposed at least once in two insertions. If the audience is known in people, divide it by the population.
- *Number of insertions*: insertions (one value per vehicle) in all functions; the formulas use n_i for a vehicle and N for the total. The column of vehicles_data in `calc_canex()` may also be named k , Danaher's notation. The argument k of `calc_agostini_duplication()` is something else: Agostini's duplication coefficient.
- *Duplication between two vehicles*: `duplication_matrix`, in people (`calc_agostini_duplication()`, `calc_hofmans_duplication()`, `calc_metheringham()`); duplications, as proportions of the population (`calc_canex()` and the sequential models). The different names signal the different units.
- *Reach* (also net audience) counts each person once; *gross audience* (also duplicated audience) is the sum of the audiences and counts a person once per audience, as in impressions.
- *Effective frequency*: `effective_frequency` in `optimize_media_plan()` and `frequency` in `calibrate_bbd()`; effective reach is the proportion exposed that many times or more.
- *Units*: every argument documents its unit. A *proportion* is a number between 0 and 1 (0.35 for 35%, not 35); percentages appear only in result columns named `percent`; counts of people appear in arguments described as "in people" and in result columns named `people`.

Assumptions of the models

Each model rests on assumptions about four things. They are the first thing to check when choosing a model.

- *Individuals*: homogeneous (every person has the same probability of being exposed to an insertion of a vehicle) or heterogeneous (probabilities differ between people, as between loyal and occasional readers). Heterogeneity makes repeated exposures concentrate in the same people, so reach grows more slowly with the number of insertions. `calc_binomial()` and `calc_sainsbury()` assume homogeneous individuals; the Beta-Binomial family (`calc_beta_binomial()`, `calc_metheringham()`, `calc_canex()`, `calc_csd()`, `calc_msad()`, `calc_cbd()`, `calc_mbd()`) lets the probability vary between people.
- *Vehicles*: homogeneous (every vehicle has the plan's average audience) or heterogeneous (each has its own audience). `calc_binomial()` and `calc_metheringham()` treat vehicles as homogeneous; `calc_sainsbury()` and the multivariate models keep each vehicle's audience.
- *Duplication between vehicles*: random (people choose vehicles independently, so the audience shared by two vehicles is the product of their audiences) or observed (taken from data). `calc_binomial()` and `calc_sainsbury()` assume random duplication; the other models with several vehicles use observed duplication.
- *Accumulation in the same vehicle*: random (a repeat insertion in a vehicle is as independent as an insertion in another vehicle) or observed through the reach after two insertions (R2), which captures that those who saw the first insertion tend to see the next. `calc_binomial()` and `calc_sainsbury()` assume random accumulation; the models for several insertions in a vehicle use the observed accumulation: R2 or, in `calc_metheringham()`, the audience duplicated between two insertions of the same vehicle.

All the models also assume that exposure probabilities are stationary (they do not change over the period of the plan). Each help page states the assumptions of its model.

Planning workflow

`media_plan()` creates a validated plan, `plan_metrics()` and `audience_metrics()` compute its metrics, `estimate_reach()` and `compare_reach_models()` estimate reach and the exposure distribution, and `optimize_media_plan()` allocates insertions under a budget.

Reach and exposure-distribution models

Plans with one or more insertions in each of several vehicles, with random duplication and accumulation: `calc_sainsbury()`, `calc_binomial()`. One vehicle with several insertions (accumulation models): `calc_beta_binomial()`, `calc_hofmans_accumulation()`. Several vehicles with one insertion each: `calc_agostini_duplication()`, `calc_hofmans_duplication()`. Several vehicles with several insertions: `calc_metheringham()`, `calc_canex()`, `calc_csd()`, `calc_msad()`, `calc_cbd()` and `calc_mbd()`. Fitting and validation: `fit_bbd_to_reach()`, `calibrate_bbd()`, `fit_nbd_exposure()`, `nbd_exposure_distribution()` and `evaluate_exposure_model()`.

Author(s)

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References

The models follow the primary sources cited in each function's help page. Development version and issue tracker: <https://github.com/majesus/mediaPlanR>.

See Also

Useful links:

- <https://github.com/majesus/mediaPlanR>
- Report bugs at <https://github.com/majesus/mediaPlanR/issues>

media_plan	<i>Create a validated cross-media plan</i>
------------	--

Description

`media_plan()` is the main data contract of `mediaPlanR`. It stores every quantity in explicit units and keeps channel-level inputs separate from derived metrics.

Usage

```
media_plan(
  data,
  population,
  channel = "channel",
  audience = "audience",
  insertions = "insertions",
  cost_per_insertion = "cost_per_insertion",
  target_audience = NULL,
  currency = "EUR"
)
```

Arguments

<code>data</code>	A data frame with one row per channel or vehicle.
<code>population</code>	Positive size of the planning universe, in people.
<code>channel, audience, insertions, cost_per_insertion</code>	Names of the columns of data that hold the channel label, the audience in people per insertion, the number of planned insertions (a non-negative integer) and the cost in currency units per insertion. Channel labels must be non-empty and unique.
<code>target_audience</code>	Optional name of a column of data with the people in the target audience per insertion. It cannot exceed audience.
<code>currency</code>	Currency label used in printed output. It does not affect any calculation.

Value

An object of class `media_plan`: a list with the validated channel table data (columns `channel`, `audience`, `insertions`, `cost_per_insertion` and, if supplied, `target_audience`), the population and the currency.

See Also

`plan_metrics()`, `estimate_reach()` and `optimize_media_plan()` take a `media_plan` object.

Examples

```
plan <- media_plan(
  data.frame(
    channel = c("TV", "Radio", "Digital"),
    audience = c(300000, 180000, 120000),
    insertions = c(4, 6, 10),
    cost_per_insertion = c(18000, 3500, 1200),
    target_audience = c(180000, 90000, 84000)
  ),
  population = 1000000,
  target_audience = "target_audience"
)
plan
```

metherringham_example *Illustrative inputs for the Metherringham model*

Description

List of arguments ready for `do.call()` with `calc_metherringham()`: three vehicles with four, three and five insertions. Original illustrative data, not derived from any published source.

Usage

```
metherringham_example
```

Format

A list with the components:

audiences Numeric vector with the audience of each vehicle, in people per insertion.

insertions Numeric vector with the number of insertions in each vehicle.

duplication_matrix Symmetric matrix, in people: the people exposed to both one insertion in vehicle *i* and one in vehicle *j* (off-diagonal), and the people exposed to two insertions of the same vehicle (diagonal, as defined in `calc_metherringham()`).

population Population size, in people.

Examples

```
data(metherringham_example)
do.call(calc_metherringham, metherringham_example)
```

msad_example

Illustrative inputs for the MSAD model

Description

List of arguments ready for `do.call(calc_msad, msad_example)`: a small two-vehicle scenario. Original illustrative data, not derived from any published source. For a dataset that reproduces published inputs for literature validation, see [msad_kim2005](#).

Usage

```
msad_example
```

Format

A list with the components:

vehicles_data Data frame with columns insertions, R1 and R2.

duplications Symmetric matrix of proportions of the population exposed to both vehicles of each pair (one insertion in each).

aggregation_order The order of aggregation, 1 : 2.

See Also

[calc_msad\(\)](#), [msad_kim2005](#)

Examples

```
data(msad_example)
do.call(calc_msad, msad_example)
```

msad_kim2005

Published inputs for the MSAD model (Kim, 2005)

Description

Minimal factual inputs (insertion counts, reach and pairwise duplication figures) reproduced from Kim (2005), included solely so that users can verify calculations against the source. They are bare numeric values reused for validation, not a reproduction of the dissertation. Kim publishes these as the inputs of the CSD example and does not publish an MSAD distribution for them; this dataset is therefore an input benchmark, and the MSAD output it produces is a derived calculation, not a published result. The audiences and reaches originate from comScore Media Metrix data of September 2003, which Kim (2005) uses and Cheong, Leckenby and Eakin (2011, Table 5) also publish.

Usage

```
msad_kim2005
```

Format

A list with the components:

vehicles_data Data frame with columns insertions, R1 and R2 for three vehicles.

duplications Symmetric matrix of proportions of the population exposed to both vehicles of each pair (one insertion in each).

aggregation_order The order of aggregation, 1:3.

References

Kim, H. G. (2005). A Canonical Sequential Aggregation Media Model. Doctoral dissertation, The University of Texas at Austin, pp. 65-71 and 80-97.

See Also

[calc_msad\(\)](#), [msad_example](#), [csd_kim2005](#)

Examples

```
data(msad_kim2005)
do.call(calc_msad, msad_kim2005)
```

nbd_exposure_distribution

Negative-Binomial exposure-count approximation

Description

Builds a Poisson-Gamma (Negative-Binomial) distribution for exposure counts. This is an unbounded count-process approximation intended for page views, ad-server exposures, or other continuous opportunity processes. It is not a finite-insertion reach model and does not represent cross-vehicle dependence.

Usage

```
nbd_exposure_distribution(
  mean_contacts,
  size,
  report_max = NULL,
  opportunities = NULL,
  tail_tolerance = 1e-06
)
```

Arguments

mean_contacts	Positive expected exposures per person.
size	Positive Gamma heterogeneity parameter. Smaller values imply stronger heterogeneity. Inf gives the Poisson limit.
report_max	Positive integer at which the reported table becomes an open tail (report_max or more), not an exact final count.

- `opportunities` Optional finite number of exposure opportunities. It is used only to diagnose the probability that the unbounded model assigns to impossible counts above that number.
- `tail_tolerance` Non-negative diagnostic tolerance.

Details

The model assumes an individual Poisson exposure process with a Gamma- distributed rate across people. Its variance is $\mu + \mu^2/size$. Because its support is unbounded, the final table row is explicitly labeled as an open tail. No tail mass is reassigned to an impossible exact exposure count.

Danaher (2007) provides direct media-planning support for Negative-Binomial page-view models and develops the multivariate extension needed to represent dependence across websites. This function is deliberately univariate and therefore must not be interpreted as Danaher's multivariate model.

Value

An `nbd_exposure` object with the open-tail distribution, reach, conditional average frequency, and scope diagnostics.

References

- Danaher, P. J. (2007). Modeling Page Views Across Multiple Websites with an Application to Internet Reach and Frequency Prediction. *Marketing Science*, 26(3), 422-437. doi:10.1287/mksc.1060.0226
- Ehrenberg, A. S. C. (1959). The Pattern of Consumer Purchases. *Applied Statistics*, 8(1), 26-41. doi:10.2307/2985810

Examples

```
model <- nbd_exposure_distribution(
  mean_contacts = 2.5, size = 1.7, report_max = 10
)
model$reach
```

optimize_media_plan	<i>Optimize a cross-media allocation</i>
---------------------	--

Description

Finds an allocation of insertions across channels under a budget, using either exhaustive search (with a verifiable global optimum) or an explicitly labeled greedy heuristic. The objective can maximize effective reach or minimize spend for a required effective reach.

Usage

```
optimize_media_plan(
  plan,
  budget,
  objective = c("max_reach", "min_cost"),
  target_reach = NULL,
```

```

    effective_frequency = 1L,
    max_insertions = plan$data$insertions,
    model = c("sainsbury", "binomial"),
    method = c("auto", "exact", "greedy"),
    max_combinations = 1e+06
  )

```

Arguments

plan	A media_plan object. Its current insertions are the default upper bounds of the search.
budget	Maximum total spend, in the plan's currency.
objective	"max_reach" maximizes effective reach within the budget (despite its name, the quantity maximized is the effective reach defined by effective_frequency; with the default effective_frequency = 1 it is the ordinary reach). "min_cost" minimizes spend, within the budget, subject to effective reach being at least target_reach.
target_reach	Required effective reach, as a proportion of the population between zero and one (0.40 for 40%, not 40), for objective = "min_cost".
effective_frequency	Effective frequency: the minimum number of exposures, f , that a person needs to count as effectively reached (a positive integer, default 1). Effective reach is the proportion of the population exposed effective_frequency times or more, that is, f or more times; it is not the proportion exposed exactly f times.
max_insertions	Integer vector with the maximum number of insertions allowed in each channel, in the order of the rows of the plan.
model	"sainsbury" or "binomial"; see estimate_reach() . Only these two models can be evaluated for every candidate allocation, because they need only the audiences and the insertions that the plan contains; the models that need observed duplications or the reach after two insertions would require those data for every possible allocation. Candidate allocations also routinely place several insertions in the same vehicle, so the Negative-Binomial approximation, which is scoped to continuous exposure processes and not to finite schedules, is not offered.
method	"exact", "greedy" or "auto", which uses exhaustive search when the number of allocations does not exceed max_combinations and the greedy heuristic otherwise.
max_combinations	Maximum number of allocations allowed for exhaustive search. The number of allocations is the product, over the channels, of one plus the maximum insertions of the channel (prod(max_insertions + 1)).

Details

Exhaustive search evaluates every allocation within max_insertions whose spend does not exceed the budget and returns the best one, so its result is a global optimum for the chosen reach model and within those bounds (the optimum is relative to the model: a different model can rank allocations differently). Ties in effective reach are broken in favor of lower spend and then higher reach. For objective = "min_cost" the cheapest allocation that reaches target_reach is returned, and an error is raised when none does.

The greedy heuristic repeatedly adds the move with the largest gain in effective reach per unit of cost. A move adds between one and effective_frequency insertions to one channel, because

fewer insertions than the effective frequency cannot by themselves create effective reach. The result is not guaranteed to be optimal and is always reported as a heuristic. With objective = "min_cost", a warning is issued if the target cannot be reached within the budget.

Value

A media_optimization object: a list with the optimized plan, the named allocation, the reach result of `estimate_reach()`, the plan metrics, the effective_frequency and effective_reach (a proportion of the population exposed effective_frequency times or more), spend, budget, target_reach and target_met, the method used, global_optimum (TRUE only for exhaustive search), combinations_evaluated and, for exhaustive search, a search_table with the spend, reach and effective reach of each evaluated allocation.

References

Aldás Manzano, J. (1998). Modelos de determinación de la cobertura y la distribución de contactos en la planificación de medios publicitarios impresos (Models for determining reach and exposure distribution in print media planning). Doctoral dissertation, Universidad de Valencia, Spain. Sections 3.3.1.1 and 3.3.1.2, the reach models this function evaluates.

See Also

`media_plan()` and `estimate_reach()`.

Examples

```
plan <- media_plan(
  data.frame(channel = c("TV", "Radio", "Digital"),
             audience = c(300000, 180000, 120000),
             insertions = c(4, 6, 10),
             cost_per_insertion = c(18000, 3500, 1200)),
  population = 1000000
)
optimized <- optimize_media_plan(
  plan, budget = 60000, objective = "max_reach",
  effective_frequency = 2, max_insertions = c(4, 8, 12)
)
optimized
optimized$global_optimum

# Cheapest allocation reaching 40% effective reach
optimize_media_plan(
  plan, budget = 60000, objective = "min_cost", target_reach = 0.40,
  effective_frequency = 2, max_insertions = c(4, 8, 12)
)$allocation
```

plan_metrics

Media-plan metrics

Description

Computes impressions, spend, rating points and cost ratios for every channel and for the whole plan, with an explicit denominator for each quantity.

Usage

```
plan_metrics(plan, reach = NULL)
```

Arguments

plan	A media_plan object.
reach	Optional unique reach of the plan, in people (for example the people element of the reach returned by <code>estimate_reach()</code>). When supplied, average frequency and cost per thousand people reached are added to the plan totals.

Details

For each channel, impressions is audience * insertions, spend is cost_per_insertion * insertions, rating_points (gross rating points) is impressions / population * 100, grp_share is the channel's share of the plan's rating points, cpm_impressions is the cost per thousand impressions and cost_per_rating_point is spend / rating_points. When the plan has a target audience, target_impressions and target_composition (target_audience / audience) are added.

Value

A media_plan_metrics object: a list with by_channel (a data frame of channel-level metrics), totals (a list with impressions, spend, grps, cpm_impressions and, when reach is supplied, reach, reach_percent, average_frequency and cost_per_thousand_reached) and the currency.

See Also

`media_plan()`, `estimate_reach()`

Examples

```
plan <- media_plan(
  data.frame(channel = c("TV", "Radio"), audience = c(300000, 180000),
             insertions = c(4, 6), cost_per_insertion = c(18000, 3500)),
  population = 1000000
)
plan_metrics(plan)

# Add reach-based metrics
plan_metrics(plan, reach = estimate_reach(plan)$reach$people)$totals
```

```
print.reach_cbd
```

Print a sequential-aggregation result

Description

By default, the full report used by the classical models: headline metrics, model parameters, the vehicles and aggregation steps, the exposure distribution, the cumulative distribution (the share of the population exposed N times or more, for each N), and numerical diagnostics. With `full = FALSE`, a compact summary: reach, average frequency, aggregation order and numerical diagnostics.

Usage

```
## S3 method for class 'reach_cbd'
print(x, full = TRUE, max_rows = 30L, ...)

## S3 method for class 'reach_csd'
print(x, full = TRUE, max_rows = 30L, ...)

## S3 method for class 'reach_mbd'
print(x, full = TRUE, max_rows = 30L, ...)

## S3 method for class 'reach_msad'
print(x, full = TRUE, max_rows = 30L, ...)
```

Arguments

x	A result of <code>calc_csd()</code> , <code>calc_msad()</code> , <code>calc_cbd()</code> or <code>calc_mbd()</code> .
full	Logical. TRUE (default) prints the full report; FALSE prints the compact summary.
max_rows	Maximum number of exposure levels listed in the full report; longer distributions are cut with a note. Use Inf to list all of them.
...	Unused.

Details

Every result class has its own print method, because the objects are long lists and each model reports different intermediate results: CSD and MSAD list the vehicles and the aggregation steps (CSD also the canonical correlations between vehicles), CBD lists how each vehicle is modeled and the canonical zero probability, and MBD lists the peeling steps. The four methods are documented together on this page.

Value

x, invisibly.

Examples

```
data(csd_kim2005)
csd <- do.call(calc_csd, csd_kim2005)
csd
print(csd, full = FALSE)
```

ratings_example

Illustrative inputs for the Sainsbury and Binomial models

Description

List of arguments ready for `do.call()` with `calc_sainsbury()` and `calc_binomial()`, the two models that assume random duplication and random accumulation. Original illustrative data, not derived from any published source.

Usage

`ratings_example`

Format

A list with the components:

audiences Numeric vector with the audience of each of three vehicles, in people per insertion.

population Population size, in people.

Examples

```
data(ratings_example)
do.call(calc_sainsbury, ratings_example)
do.call(calc_binomial, ratings_example)
```

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